

The Poole Local-Geometry Program

Hysteretic Surface Selection, Defect Calculus, Channel Spectrometry, Signed Membrane Basins,
and the Defect–Channel Bridge

Rooke Alan Poole

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Abstract

This manuscript presents the Poole Local-Geometry Program as a finite mathematical and computational theory of local support geometry on three-dimensional Moore-neighborhood lattices. The core rule is the hysteretic update

$$P(x)_i = \mathbf{1}\{5 \leq N_i(x) \leq 7 + 2x(i)\},$$

which is exactly B5–7/S5–9. From this single local rule the manuscript develops a theorem spine for surface selection, stable one-cell sheets, exact first-step planar defect response, rectangle and channel laws, selected non-planar first-response subfamilies, and the defect–channel bridge. The PooleQ/P layer preserves birth, survival, high-support, and strain channels as a geometry spectrometer for measured classical binary fields. The signed-membrane layer extends the same support-window logic to opposite-signed populations and benchmarks when preparation induces thin, low-imbalance reaction-interface basins. The bridge benchmark then tests whether pre-warmup channel spectra predict signed basin quality beyond density and mirror-symmetry baselines.

The claim boundary is deliberately strict. The manuscript proves finite local theorems where local count arguments close; it reports verifier and benchmark results where scripts and finite experiments support a claim; it leaves atlas observations, universal out-of-family prediction, continuum interpretation, and physical interpretation as open material. No continuum, cosmological, propulsion, energy, quantum-state-copying, or hardware claim is established by the results here.

Submission scope note

This manuscript is self-contained. The relevant definitions, theorems, proof capsules, verifier tables, benchmark tables, claim lanes, and open problems are integrated directly here. External citations are limited to outside literature needed for background boundaries such as no-cloning, p-bit terminology, surface-code syndrome context, and antimatter experimental scope. Reproducibility files and benchmark data are packaged as ancillary material rather than as citations to earlier author manuscripts.

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Part I

Program, scope, and local rule

Chapter 1

Research program and claim lanes

1.1 What the program studies

The Poole Local-Geometry Program studies deterministic and stochastic consequences of local support thresholds on a three-dimensional cubic lattice. The center of the program is not an appeal to continuum physics or hardware speculation. It is the finite combinatorics of a local automaton and the benchmarkable morphology that grows from that automaton.

The program has four active layers.

1. **Defect Calculus:** the theorem layer. It proves how the B5–7/S5–9 rule selects sheets and responds to planar defects.
2. **PooleQ/P Capture:** the channel-spectrometry layer. It preserves the local event channels that the raw update normally collapses.
3. **Signed Membranes:** the opposite-sign basin layer. It asks when signed populations form stable, thin, low-imbalance reaction interfaces.
4. **Bridge Benchmark:** the cross-layer test. It asks whether PooleQ/P channel spectra predict signed basin entry beyond density or mirror symmetry alone.

1.2 Claim lanes

Every statement in the manuscript is assigned to one of five lanes.

Theorem-level. A finite combinatorial consequence of the stated local rule.

Verifier-level. A result reproduced by an exact script or reference checker.

Benchmark-level. A finite experiment with seeds, controls, summary statistics, and baselines.

Empirical-atlas level. A reproducible numerical pattern used to guide theorem discovery, not itself a proof.

Bridge/open level. A bounded interpretation or target theorem that links layers but is not yet a closed theorem.

This lane discipline is part of the scientific contribution. It prevents exact finite theorems, finite benchmarks, and speculative interpretations from being mixed into one unsupported claim.

1.3 Boundary statement

This manuscript does not claim a physical theory of spacetime, a method for copying or reconstructing unknown quantum amplitudes, an antimatter production or propulsion technology, or a universal controller. The quantum-state boundary is consistent with the standard no-cloning results of Wootters–Zurek and Dieks [1, 2]. The established object is narrower: a local-geometry theory and benchmark suite for Moore-neighborhood support rules, channel spectra, and signed reaction-interface basins.

Chapter 2

The finite Poole lattice

Definition 2.1 (Periodic cubic lattice). *Fix $L \in \mathbb{N}$. The finite lattice is*

$$\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3.$$

A binary configuration is a map $x : \Lambda_L \rightarrow \{0, 1\}$.

Definition 2.2 (Moore neighborhood). *The radius-one Moore neighborhood in three dimensions is*

$$N_{26} = \{-1, 0, 1\}^3 \setminus \{(0, 0, 0)\}.$$

The active neighbor count at site i is

$$N_i(x) = \sum_{\delta \in N_{26}} x(i + \delta),$$

where addition is modulo L in each coordinate.

Definition 2.3 (Base Poole map). *The base Poole update is*

$$P(x)_i = \mathbf{1}\{5 \leq N_i(x) \leq 7 + 2x(i)\}. \quad (2.1)$$

The evolution is $x_{t+1} = P(x_t)$.

Proposition 2.4 (Equivalence to B5–7/S5–9). *The compact law (2.1) is exactly the birth-survival rule B5–7/S5–9.*

Proof. If $x(i) = 0$, the upper bound in (2.1) is 7, so the site is born exactly for $N_i \in \{5, 6, 7\}$. If $x(i) = 1$, the upper bound is 9, so the site survives exactly for $N_i \in \{5, 6, 7, 8, 9\}$. $\square$

2.1 Normal forms

The same rule will be used in several equivalent forms.

Birth-survival form.

$$P(x)_i = (1 - x(i))\mathbf{1}\{N_i \in \{5, 6, 7\}\} + x(i)\mathbf{1}\{N_i \in \{5, 6, 7, 8, 9\}\}.$$

Capacity form. Let $C_i(x) = 7 + 2x(i)$. Then $P(x)_i = 1$ iff $5 \leq N_i(x) \leq C_i(x)$.

Deficit-excess form. Let

$$D_i = (5 - N_i)_+, \quad E_i = (N_i - C_i)_+, \quad \psi_i = E_i - D_i.$$

Then $P(x)_i = 1$ iff $D_i = E_i = 0$, equivalently $\psi_i = 0$.

Channel form. The accepted birth channels are B_5, B_6, B_7 and the accepted survival channels are S_5, S_6, S_7, S_8, S_9 .

2.2 Why the $+2x(i)$ term matters

The term $2x(i)$ is not an added field. It is a local hysteretic allowance: active cells may survive two counts above the inactive birth ceiling. This difference is exactly what allows a one-cell sheet to persist. Without it, a full coordinate plane has $N_i = 8$ at each active site and dies under the memory-reduced rule B5-7/S5-7.

Part II

Hysteretic surface selection and Defect Calculus

Chapter 3

Basic structural theorems

Proposition 3.1 (Cubic equivariance). *The map P commutes with translations of Λ_L , coordinate permutations, and coordinate sign flips.*

Proof. Each operation only permutes the elements of the Moore neighborhood and therefore preserves the neighbor count $N_i(x)$. $\square$

Proposition 3.2 (Finite propagation speed). *If two initial configurations agree on the radius- t Moore ball around a site i , then their states at i agree after t updates.*

Proof. One update depends only on a radius-one Moore ball. Iterating the dependency gives a radius- t dependency after t updates. $\square$

Proposition 3.3 (Eventual periodicity). *For fixed L , every trajectory under P is eventually periodic with preperiod plus period at most 2^{L^3} .*

Proof. There are only 2^{L^3} configurations. A deterministic map on a finite set must eventually revisit a previous state. $\square$

Proposition 3.4 (Non-monotonicity). *The Poole map is not monotone with respect to coordinatewise order.*

Proof. Let $x \leq y$ and choose an inactive target site i . Give x exactly five active neighbors of i . Let y add three more active neighbors, so $N_i(y) = 8$. Then $P(x)_i = 1$ but $P(y)_i = 0$, because inactive cells are over-supported above 7. $\square$

Proposition 3.5 (Non-reversibility). *For $L \geq 3$, P is not injective and hence not reversible.*

Proof. The all-zero configuration maps to itself. The all-one configuration has $N_i = 26$ at every site, so every active cell is over-supported and dies. Thus two distinct configurations map to the all-zero configuration. $\square$

Chapter 4

Surface selector theorem

Definition 4.1 (Coordinate-flat local pattern). *A configuration is locally a coordinate r -flat through a site if, in the centered $3 \times 3 \times 3$ Moore block, the active sites are exactly the sites of an r -dimensional coordinate subspace through that site.*

Lemma 4.2 (Flat local degree count). *A site in a complete coordinate r -flat has local active neighbor count*

$$3^r - 1.$$

Thus the ideal counts for point, line, sheet, and filled bulk are 0, 2, 8, 26.

Proof. The centered block contains 3^r sites of the coordinate r -flat, one of which is the center. The rest are active neighbors. $\square$

Theorem 4.3 (Local surface selector). *Among ideal coordinate-flat local bulk types, the B5-7/S5-9 survival window preserves exactly the locally two-dimensional sheet type.*

Proof. The survival window is $5 \leq N_i \leq 9$. Of the counts 0, 2, 8, 26, only 8 lies in this interval. $\square$

Theorem 4.4 (Stable active plane). *Assume $L \geq 3$. A full coordinate plane is a fixed point of the base Poole map.*

Proof. Each active site in the plane has the other eight sites in its in-plane 3×3 footprint as active neighbors, so $N_i = 8$ and it survives. A site one layer above or below sees a full 3×3 footprint with $N_i = 9$, which is too high for birth. Sites farther away see zero active neighbors. The configuration is unchanged. $\square$

Corollary 4.5 (Sheet memory). *Under the memory-reduced ablation B5-7/S5-7, the full active plane dies in one step because its active cells have $N_i = 8$.*

Chapter 5

Planar defect calculus

Definition 5.1 (Defective sheet). *Identify a stable coordinate sheet with the two-dimensional torus $T_L = (\mathbb{Z}/L\mathbb{Z})^2$. Let $D \subset T_L$ be a defect mask of missing sheet sites. The defective-sheet configuration has $x_D(u, c) = 1$ for $u \notin D$ and all non-sheet layers inactive.*

Definition 5.2 (Open and closed defect counts). *For $u \in T_L$, define*

$$q_D(u) = \sum_{\eta \in \{-1,0,1\}^2 \setminus \{0\}} \mathbf{1}\{u + \eta \in D\},$$

$$r_D(u) = \sum_{\eta \in \{-1,0,1\}^2} \mathbf{1}\{u + \eta \in D\}.$$

The count q_D is the open in-plane defect count. The count r_D is the closed 3×3 defect-footprint count.

Theorem 5.3 (One-step planar defect selector). *For the base Poole rule, the first update of a defective sheet x_D is determined exactly as follows.*

1. *An active sheet site $u \notin D$ survives exactly when $q_D(u) \leq 3$.*
2. *An inactive in-plane defect site $u \in D$ is born exactly when $1 \leq q_D(u) \leq 3$.*
3. *A normal-layer site at $(u, c+1)$ or $(u, c-1)$ is born exactly when $2 \leq r_D(u) \leq 4$.*
4. *All layers with $|z - c| \geq 2$ remain inactive after the first update.*

Proof. An active or inactive site in the sheet plane sees only the eight in-plane neighbors, so the raw count is $8 - q_D(u)$. Active survival requires $5 \leq 8 - q_D(u) \leq 9$, equivalent to $q_D(u) \leq 3$. Inactive birth requires $5 \leq 8 - q_D(u) \leq 7$, equivalent to $1 \leq q_D(u) \leq 3$. A normal-layer site sees a 3×3 sheet footprint, so the count is $9 - r_D(u)$; birth requires $5 \leq 9 - r_D(u) \leq 7$, equivalent to $2 \leq r_D(u) \leq 4$. Farther layers see no active neighbors during the first update. $\square$

Corollary 5.4 (Stable pinhole gas). *If every closed 3×3 footprint contains at most one defect site, then the defective sheet is fixed. In particular, a single one-cell pore is exactly stable.*

5.1 Operator table

Cell class	Raw count	Rule test	Result
Active sheet site $u \notin D$	$8 - q_D(u)$	survival	survives iff $q_D \leq 3$
Missing in-plane defect $u \in D$	$8 - q_D(u)$	birth	born iff $1 \leq q_D \leq 3$
Normal-layer site $(u, c \pm 1)$	$9 - r_D(u)$	birth	born iff $2 \leq r_D \leq 4$
Distance ≥ 2 layer	0	birth	inactive

Chapter 6

Rectangle laws and reconstruction boundary

The front-manuscript proof capsules and ancillary verifier tables contain the rectangle and line-hole proofs, prime-correction model, work laws, and reconstruction audit. The central closed family is recorded here because it is a major bridge from raw defect geometry to channel spectra.

Theorem 6.1 (Rectangle first-step zero-death family). *For rectangular planar holes in the theorem-level subfamily, the first-step response is a closed zero-death response with raw birth count*

$$B_0(a, b) = 4a + 4b + 12 = 2P + 12,$$

where $P = 2a + 2b$ is the rectangle perimeter.

Theorem 6.2 (Rectangle channel spectrum). *For the same rectangle family, the first-step birth-channel spectrum is*

$$(N_5, N_6, N_7) = (12, 4a + 4b - 16, 16).$$

Theorem 6.3 (Prime-corrected rectangle response). *Under the default prime-sharpened model that removes the raw N_7 birth group, the rectangle response is*

$$B_\varphi(a, b) = 4a + 4b - 4.$$

Remark 6.4 (Model status). *The raw rectangle laws are theorem-level consequences of the first-step selector. The prime-sharpened response is a model-tagged correction, not a claim that the base automaton secretly removes N_7 births. This distinction must remain visible in the final manuscript.*

6.1 Q/P-style rectangle reconstruction boundary

The rectangle reconstruction results are bounded by their prior: they apply under a rectangle hypothesis and channel-resolved readout. They should not be promoted into arbitrary-shape reconstruction until a separate theorem or benchmark supports that extension.

Chapter 7

Defect verifier import v1.1

This chapter moves the Defect Calculus verifier layer into the front manuscript. The imported verifier is not used as a substitute for the preceding proofs. Its role is implementation hardening: it checks indexing, channel separation, periodic-boundary behavior, model tags, and reconstruction formulas against explicit finite updates.

7.1 Closed-layer pass/fail summary

The regenerated v1.1 verifier reproduces the closed first-step audit families from the Defect Calculus layer. The empirical multi-step traces are included as ancillary reference data only and are not counted as theorem-level pass/fail evidence.

Table 7.1: v1.1 regenerated Defect verifier pass/fail summary.

Metric	Value
Rectangle cases	841
Rectangle raw-law matches	841
Rectangle channel-spectrum matches	841
Rectangle prime-response matches	841
Rectangle overcompensation matches	841
Line-hole cases	80
Line-hole raw-law matches	80
Line-hole prime-response matches	80
Random planar selector cases	720
Random planar selector matches	720
Random planar nonlocal events total	0
Q/P rectangle inversion cases	1225
Q/P raw inversion matches	1225
Q/P prime inversion matches	1225
Empirical timeseries rows	18
All closed-layer checks passed	True

The equality of actual and predicted event sets in the random planar selector audit is the most important implementation check. It tests arbitrary finite masks rather than only rectangles and confirms that the first-step event set is completely accounted for by the two planar counts q_D and r_D .

7.2 Rectangle, channel, prime, and overcompensation audit

For every $2 \leq a, b \leq 30$, the verifier constructs a stable sheet, removes an $a \times b$ rectangle, performs one raw update, and records births, deaths, active counts, and birth-channel counts. Every row satisfies

$$B_0(a, b) = 4a + 4b + 12, \quad D_0(a, b) = 0,$$

$$(N_5, N_6, N_7) = (12, 4a + 4b - 16, 16),$$

$$B_\phi(a, b) = N_5 + N_6 = 4a + 4b - 4,$$

and the first-step overcompensation criterion

$$|A_1| > |A_{\text{sheet}}| \iff (a - 4)(b - 4) < 28.$$

Table 7.2: Representative v1.1 rectangle/channel/prime audit rows.

a	b	B_0	pred.	D	N_5	N_6	N_7	B_ϕ	pred.	ok
2	2	28	28	0	12	0	16	12	12	True
2	3	32	32	0	12	4	16	16	16	True
2	10	60	60	0	12	32	16	44	44	True
2	30	140	140	0	12	112	16	124	124	True
10	10	92	92	0	12	64	16	76	76	True
10	20	132	132	0	12	104	16	116	116	True
10	30	172	172	0	12	144	16	156	156	True
30	30	252	252	0	12	224	16	236	236	True

Figure omitted in single-file source.
The surrounding caption, tables, and ancillary data contain the reported result.

Figure 7.1: Regenerated rectangle raw first-step birth surface for $2 \leq a, b \leq 30$.

Figure omitted in single-file source.
The surrounding caption, tables, and ancillary data contain the reported result.

Figure 7.2: Regenerated rectangle birth-channel spectrum slice. The N_5 and N_7 groups are constant while N_6 carries the perimeter-dependent term.

Figure omitted in single-file source.
The surrounding caption, tables, and ancillary data contain the reported result.

Figure 7.3: Regenerated first-step overcompensation region. The criterion is $(a - 4)(b - 4) < 28$.

7.3 Line-hole edge audit

The one-cell-wide family is a degenerate boundary case and is not folded into the $a, b \geq 2$ rectangle law. The regenerated audit covers $1 \leq n \leq 80$ and checks

$$B_0(1) = B_\phi(1) = 0,$$

while for $n \geq 2$ it checks

$$B_0(1, n) = 7n, \quad B_\phi(1, n) = 7n - 14.$$

Table 7.3: Representative v1.1 line-hole edge audit rows.

n	B_0	pred.	D	N_5	N_6	N_7	B_ϕ	pred.	ok
1	0	0	0	0	0	0	0	0	True
2	14	14	0	0	0	14	0	0	True
3	21	21	0	0	7	14	7	7	True
10	70	70	0	0	56	14	56	56	True
40	280	280	0	0	266	14	266	266	True
80	560	560	0	0	546	14	546	546	True

Figure omitted in single-file source.

The surrounding caption, tables, and ancillary data contain the reported result.

Figure 7.4: Regenerated raw and default prime-corrected line-hole response for $1 \leq n \leq 80$.

7.4 Arbitrary planar selector audit

For each random finite planar mask D , the verifier removes D from a stable sheet, performs one raw update, and compares the full event set against the q_D/r_D prediction:

in-plane births: $u \in D, \quad 1 \leq q_D(u) \leq 3,$

deaths: $u \notin D, \quad q_D(u) \geq 4,$

normal births: $z = c \pm 1, \quad 2 \leq r_D(u) \leq 4.$

All 720 generated rows match, and the total number of nonlocal first-step events is zero.

Table 7.4: First ten v1.1 random planar selector audit rows.

case	box	removed	actual E	pred. E	plane B	plane D	adj. B	ok
1	7x12	27	185	185	24	5	156	True
2	6x4	1	0	0	0	0	0	True
3	5x11	7	30	30	4	0	26	True
4	12x7	44	150	150	17	15	118	True
5	8x9	30	185	185	27	14	144	True
6	8x10	43	151	151	22	19	110	True
7	12x4	16	69	69	9	6	54	True
8	10x10	22	154	154	21	3	130	True
9	12x4	7	31	31	5	0	26	True
10	6x5	16	83	83	14	7	62	True

Figure omitted in single-file source.

The surrounding caption, tables, and ancillary data contain the reported result.

Figure 7.5: Actual first-step event counts versus q_D/r_D -predicted event counts for 720 random finite planar masks.

7.5 Q/P-style rectangle reconstruction audit

The reconstruction audit remains explicitly prior-bounded: it assumes a rectangle. With area $A = ab$ and raw response B_0 , the sum is

$$S = a + b = \frac{B_0 - 12}{4}.$$

With default prime response B_ϕ , the sum is

$$S = a + b = \frac{B_\phi + 4}{4}.$$

Then a and b are the integer roots of

$$t^2 - St + A = 0.$$

The v1.1 audit covers $2 \leq a, b \leq 36$, giving 1225 cases, and both raw and prime inversions match in every case.

Table 7.5: Representative v1.1 Q/P-style rectangle inversion rows.

a	b	A	B_0	B_ϕ	raw rec.	phi rec.	raw ok	phi ok
2	2	4	28	12	2x2	2x2	True	True
2	7	14	48	32	2x7	2x7	True	True
2	36	72	164	148	2x36	2x36	True	True
10	20	200	132	116	10x20	10x20	True	True
10	36	360	196	180	10x36	10x36	True	True
36	36	1296	300	284	36x36	36x36	True	True

7.6 Empirical multi-step trace boundary

The empirical planar relaxation trace is included because it is useful for later law discovery and for aligning the empirical atlas. It is not theorem-level evidence. The current imported trace rows record selected rectangles at $t = 0, 1, 2, 3, 6, 12$ under the raw map. They should remain in the empirical/open lane until a closed multi-step theorem is proved.

Table 7.6: Representative empirical relaxation trace rows. These rows are not theorem-level claims.

case	t	active	births	deaths	delta
2x2	0	9212	0	0	0
2x2	1	9240	28	0	28
2x2	2	9200	16	56	-40
2x2	3	9256	72	16	56
4x12	0	9168	0	0	0
4x12	1	9244	76	0	76
10x20	0	9016	0	0	0
10x20	1	9148	132	0	132

7.7 Evidence files imported in v1.1

The following generated evidence files are included under `data/defect_verifier/`: rectangle-square-channel audit, line-hole audit, arbitrary planar selector audit, Q/P rectangle inversion audit, rectangle

overcompensation audit, empirical relaxation traces, evidence summary, and checksum manifest. These files make the Defect verifier layer inspectable without relying only on printed representative rows.

Part III

PooleQ/P Capture and channel spectrometry

Chapter 8

Channel-resolved Poole transform

The ordinary update collapses all allowed births and survivals into a single binary next state. PooleQ/P keeps the local event channels separated before this collapse.

Definition 8.1 (Birth and survival channels). *For $k = 0, 1, \dots, 26$ define*

$$B_k(i; x) = (1 - x(i))\mathbf{1}\{N_i(x) = k\}, \quad S_k(i; x) = x(i)\mathbf{1}\{N_i(x) = k\}.$$

The active Poole channels are B_5, B_6, B_7 and S_5, S_6, S_7, S_8, S_9 .

Definition 8.2 (PooleQ/P feature vector). *The local channel feature vector is*

$$\Phi_i(x) = (B_5, B_6, B_7, S_5, S_6, S_7, S_8, S_9, O_{10+}, \psi)_i,$$

where

$$O_{10+}(i; x) = x(i)\mathbf{1}\{N_i(x) \geq 10\}$$

and

$$\psi_i(x) = \max\{0, N_i(x) - (7 + 2x(i))\} - \max\{0, 5 - N_i(x)\}.$$

Remark 8.3 (Readout boundary). *The channel transform is richer than the raw next-state map. If only the binary output $P(x)_i$ is observed, then channels such as B_5, B_6, B_7 are not distinguished. Exact channel claims therefore require typed readout, spatial demultiplexing, or feature extraction from the local state before channel collapse.*

Chapter 9

Probabilistic p-bit form

Let $X_j \sim \text{Bernoulli}(p_j)$ independently over sites in a local product approximation. The terminology of p-bits is used in the standard sense of controllable stochastic binary variables in probabilistic computing [3, 4]. Since the center site is not included in its own Moore count, X_i is independent of N_i under this model.

Definition 9.1 (Local generating function). *For a fixed site i define*

$$G_i(z) = \prod_{j \in N_{26}(i)} (1 - p_j + p_j z).$$

Then

$$\Pr[N_i(X) = k] = [z^k]G_i(z).$$

Proposition 9.2 (Expected Poole channels). *Under the independent Bernoulli model,*

$$\mathbb{E}[B_k(i; X)] = (1 - p_i)[z^k]G_i(z),$$

$$\mathbb{E}[S_k(i; X)] = p_i[z^k]G_i(z).$$

Therefore

$$q_i := \mathbb{E}[P(X)_i] = (1 - p_i)C_i^{5:7} + p_i C_i^{5:9},$$

where $C_i^{a:b} = \sum_{k=a}^b [z^k]G_i(z)$.

Corollary 9.3 (Hysteretic probability decomposition). *Because $C_i^{5:9} = C_i^{5:7} + C_i^{8:9}$,*

$$q_i = C_i^{5:7} + p_i C_i^{8:9}.$$

The center probability gates only the upper memory band 8, 9.

Proposition 9.4 (Differentiability). *For $\ell \in N_{26}(i)$,*

$$\frac{\partial}{\partial p_\ell} [z^k]G_i(z) = [z^k] \left((z - 1) \prod_{j \in N_{26}(i), j \neq \ell} (1 - p_j + p_j z) \right).$$

For the center site,

$$\frac{\partial q_i}{\partial p_i} = C_i^{8:9}.$$

Chapter 10

Correlation residue and geometry signatures

Given samples $X^{(1)}, \dots, X^{(M)}$, empirical channel rates can be compared with independent-site predictions built from the observed marginals. The scalar activation residue is

$$\Delta_i = \widehat{q}_i - q_i^{\text{ind}}.$$

A channel-wise residue vector records deviations in selected birth and memory-survival channels.

Definition 10.1 (Poole coherence). *A simple global structured-correlation statistic is*

$$C_P = \sum_{i \in V} \|R_i\|_2^2.$$

A large value indicates that local Poole channel rates are not explained by the product of local marginals. It does not by itself prove a quantum mechanism or a particular physical noise model.

Definition 10.2 (Geometry spectrometer). *The v5.6 signature is*

$$\mathcal{S}_P(X) = (R_B(X), R_H(X), R_\psi(X)),$$

where R_B tracks birth-window/boundary geometry, R_H tracks high-support geometry, and R_ψ tracks support-strain geometry.

10.1 Exact-cardinality footprint logic

Around a stable sheet, a normal-layer candidate sees $N = 9 - r_D(u)$. Therefore

$$B_7 \iff r_D = 2, \quad B_6 \iff r_D = 3, \quad B_5 \iff r_D = 4.$$

With selected input defects and fixed bias defects in the footprint, these channels become exact-cardinality detectors. The verifier-import chapters and ancillary tables contain the raw biased-footprint Boolean gates and adder readouts.

10.2 Syndrome-field boundary

PooleQ/P may be applied to already-measured readout strings, stochastic p-bit samples, or syndrome-event histories. This is the same measurement-side regime in which surface-code decoders operate on classical syndrome data rather than on unknown quantum amplitudes [5, 6, 7]. It does not copy, clone, or reconstruct an unknown unmeasured quantum state. It is a classical transform applied after measurement.

Part IV

Signed reaction-membrane basins

Chapter 11

Signed lattice and membrane measurements

Definition 11.1 (Signed state). *The signed membrane layer uses*

$$s_i(t) \in \{-1, 0, +1\}.$$

Positive and negative populations update by same-sign support windows, while opposite support can collapse a candidate or active cell when it exceeds a threshold k .

Definition 11.2 (Interface measurements). *Let N_i^+ and N_i^- be positive and negative local support counts. A local overlap proxy is*

$$q_i = \min(N_i^+ + \mathbf{1}_{s_i=+1}, N_i^- + \mathbf{1}_{s_i=-1}).$$

The interface support is

$$Q = \sum_{i \in \Omega} q_i \mathbf{1}_{q_i \geq q_0},$$

and the normalized density is

$$Q_\rho = Q/|\Omega|.$$

The signed imbalance is

$$\epsilon_\pm = \frac{|N^+ - N^-|}{N^+ + N^- + \varepsilon}.$$

The membrane quality proxy is

$$M_P = \frac{Q_\rho}{1 + \sigma_y}(1 - \epsilon_\pm).$$

Remark 11.3 (Proxy status). *M_P is a benchmark descriptor, not a physical thrust or energy law. Legacy names in older outputs such as `thrust_proxy` must be treated as interface-quality proxies.*

Chapter 12

Preparation beats forcing

The signed membrane benchmark tests induction, maintenance, perturbation stress, survival-window ridges, geometry/boundary robustness, and mirror-quality basin size. Its central benchmark conclusion is that stable signed membranes are induced by symmetry-preserving preparation and preserved by passive or gentle dynamics. Broad high-motion forcing degrades the induced basin.

12.1 Induction result

The original signed benchmark showed that mirrored signed initialization plus a short passive warmup produced high practical scores, high interface density, thin membranes, and good-regime rates of 100 percent, while random signed initialization and preformed sheet controls were weaker.

Table 12.1: Signed-membrane induction summary from the source benchmark.

Protocol	Practical score	Q_ρ	σ_y	Good rate
mirror+10 membrane	0.785	0.548	0.000	100.0%
mirror+20 hybrid	0.751	0.522	0.001	100.0%
mirror+10 hybrid	0.745	0.520	0.007	100.0%
mirror+0 hybrid	0.700	0.507	0.080	100.0%
random hybrid	0.492	0.450	0.648	68.8%
preformed sheet hybrid	0.254	0.406	1.713	43.8%

12.2 Maintenance result

After induction, passive membrane-lock and gentle local maintainers preserve the basin more effectively than broad static forcing.

Table 12.2: Signed-membrane maintenance summary.

Maintainer	Practical	Q_ρ	σ_y	$\epsilon_\pm$	Moves/tick
passive membrane-lock	0.785	0.548	0.000	0.003	0.000
same-move random	0.771	0.536	0.006	0.003	0.777
interface-blind	0.761	0.529	0.000	0.003	0.806
hybrid guard	0.745	0.520	0.007	0.004	2.079
focus only	0.744	0.519	0.007	0.004	2.081
static broad forcing	0.602	0.485	0.191	0.014	84.233

12.3 Rule ridge

The source signed-membrane rule sweep found a stronger ridge near B5-7/S3-11 and B5-7/S4-11 than the early canonical B5-7/S5-9 rule. This is important because it prevents the program from treating one rule as sacred. The local geometry is rule-sensitive.

Table 12.3: Source signed-membrane survival-window ridge summary.

Rule	Practical	Good rate	Q_ρ	σ_y
B5-7/S4-11/k3	0.7559	1.000	0.5255	0.0001
B5-7/S3-12/k3	0.7502	1.000	0.5109	0.0001
B5-7/S3-11/k3	0.7480	1.000	0.5285	0.0025
B5-7/S5-12/k3	0.7420	1.000	0.5069	0.0033
B5-7/S5-10/k3	0.6515	1.000	0.5163	0.1826

Part V

The defect–channel bridge and spectral-basin benchmark

Chapter 13

Exact bridge theorem

The theorem bridge is the cleanest cross-paper result. It says that the Defect Calculus and PooleQ/P layers are not merely analogous. At the first step of a defective sheet, they are the same local information represented in two coordinate systems.

Theorem 13.1 (Defect-channel identity). *Let x_D be a defective stable sheet. For the first update:*

$$\begin{aligned} u \notin D : & \quad S_{8-q_D(u)}, \\ u \in D : & \quad B_{8-q_D(u)}, \\ z = c \pm 1 : & \quad B_{9-r_D(u)}. \end{aligned}$$

Thus the q_D/r_D selector and the PooleQ/P channel spectrum are equivalent first-step local descriptions.

Proof. The planar defect selector already gives the raw counts. An active sheet site has count $8 - q_D(u)$ and therefore belongs to survival channel $S_{8-q_D(u)}$ if it survives. An inactive in-plane defect has the same raw count but is a birth candidate, so it belongs to $B_{8-q_D(u)}$ when the birth window accepts it. A normal-layer candidate has count $9 - r_D(u)$ and therefore belongs to $B_{9-r_D(u)}$ when accepted. These are exactly the channel labels of the PooleQ/P transform. $\square$

Remark 13.2 (What is exact and what is not). *The identity above is theorem-level. The spectral-basin predictor is not theorem-level. It is a benchmark hypothesis: pre-warmup channel spectra should predict later signed membrane basin quality.*

Chapter 14

Spectral-basin benchmark

For a signed initial condition $s \in \{-1, 0, +1\}^\Lambda$, the benchmark computes binary fields

$$X_+ = \mathbf{1}_{\{s=+1\}}, \quad X_- = \mathbf{1}_{\{s=-1\}}, \quad X_{abs} = \mathbf{1}_{\{|s|=1\}},$$

and an interface field detecting local plus/minus contact. It then extracts spectral features from these fields and tests whether those features predict final M_P after warmup.

14.1 Hard-mode rule-ridge result

The v1.3 hard-mode ridge run tested three rules: B5-7/S5-9/k3, B5-7/S4-11/k3, and B5-7/S3-11/k3. Structured signed preparations beat every control under every tested rule.

Table 14.1: v1.3 bridge rule-ridge separation.

Rule	Family	n	Mean M_P	Min M_P	Max M_P
B5-7/S3-11/k3	control	16	0.001585	0.000000	0.016673
B5-7/S3-11/k3	structured	32	0.227726	0.116442	0.313059
B5-7/S4-11/k3	control	24	0.001145	0.000000	0.015726
B5-7/S4-11/k3	structured	24	0.223206	0.112040	0.298879
B5-7/S5-9/k3	control	21	0.000293	0.000000	0.006150
B5-7/S5-9/k3	structured	27	0.112651	0.061871	0.142781

Across the run,

$$\max M_P(\text{control}) = 0.01667, \quad \min M_P(\text{structured}) = 0.06187.$$

The weakest structured case therefore remains above the strongest control case by about $3.71\times$.

14.2 Protocol controls

The mirrored-random control is a crucial negative control. It has high corrected mirror quality but near-zero final membrane quality. This prevents the result from collapsing into the trivial claim that mirror symmetry alone is enough.

Table 14.2: v1.3 bridge protocol summary.

Protocol	n	Mean mirror quality	Mean M_P	Max M_P
mirrored random control	17	0.754132	0.003075	0.016673
same-density random	9	0.009658	0.000000	0.000000
signed shuffle control	21	0.010783	0.000320	0.006690
single-sign sheet control	14	0.000000	0.000000	0.000000
sheet-pair aligned	20	0.955595	0.245144	0.313059
sheet-pair degraded	19	0.492711	0.160364	0.225343
sheet-pair shifted	20	0.556214	0.191917	0.273400
sheet-pair widegap	12	0.924945	0.179730	0.256944
sheet-patch pair	12	0.423170	0.145072	0.227745

14.3 Predictive models

The predictor result is useful but must be stated carefully. In all cases, spectral features beat density and mirror baselines but trail explicit protocol/rule labels and combined numeric models. In structured-only rule-ridge tests, rule-aware design features dominate the fixed spectral bridge. This is a target for v1.4 rather than a failure of the basin-entry result.

Table 14.3: Selected v1.3 bridge predictive results.

Scope	Model	Mean R^2	RMSE	MAE	Spearman
all cases	density baseline	0.144	0.0977	0.0829	0.450
all cases	mirror baseline	0.497	0.0743	0.0581	0.730
all cases	protocol one-hot	0.814	0.0449	0.0316	0.849
all cases	spectral bridge	0.768	0.0506	0.0384	0.840
all cases	combined numeric	0.869	0.0372	0.0314	0.914
structured only	density baseline	0.042	0.0623	0.0531	0.306
structured only	mirror baseline	0.038	0.0613	0.0526	0.431
structured only	protocol one-hot	0.219	0.0582	0.0503	0.513
structured only	spectral bridge	0.138	0.0600	0.0491	0.452
structured only	combined numeric	0.743	0.0325	0.0255	0.859

Figure omitted in single-file source.
The surrounding caption, tables, and ancillary data contain the reported result.

Figure 14.1: Rule-ridge bridge result: structured preparations remain separated from controls under all tested rules, and S3–11/S4–11 improve mean membrane quality over S5–9.

Figure omitted in single-file source.
The surrounding caption, tables, and ancillary data contain the reported result.

Figure 14.2: All-case prediction scatter from the v1.3 bridge run. This figure is benchmark evidence, not theorem evidence.

Chapter 15

Bounded conclusion of the bridge layer

The exact theorem is the defect–channel identity. The empirical result is basin-entry generalization: structured Poole-relevant signed geometry enters a membrane basin while high-mirror random geometry does not. The remaining open issue is fine-quality prediction inside structured families across unseen rules. A rule-aware spectral signature is the appropriate next benchmark target.

Part VI

Coverage-audited proof and benchmark expansion v1.1

Chapter 16

v1.1 coverage-audited import pass

This v1.0 pass began the coverage-audited expansion that moved proof capsules and benchmark tables into the front manuscript. In the present manuscript, the earlier source-PDF appendices are no longer used as citation targets; the integrated manuscript and ancillary data carry the record.

16.1 What v1.0 imports

The following material is imported into the front manuscript in this pass:

1. structural proof capsules for equivariance, finite propagation, eventual periodicity, non-monotonicity, non-reversibility, active-degree constraints, surface selection, stable planes, slab rejection, and memory ablation;
2. planar-defect proof capsules for the q_D/r_D selector, isolated pore stability, crack/line-source behavior, rectangle response, channel spectrum, prime-corrected response, work laws, and rectangle reconstruction boundary;
3. Q/P proof capsules for channel disjointness, probabilistic p-bit form, differentiability, correlation residue, exact-cardinality defect logic, raw footprint gates, adders, and syndrome-field boundaries;
4. full signed-membrane benchmark tables for induction, maintenance, stress, rule ridge, geometry/boundary robustness, and mirror-quality basin size;
5. bridge model-comparison, protocol-summary, and rule-ridge tables from the v1.3 hard-mode ridge benchmark;
6. a new source-to-section coverage matrix and omission checklist.

16.2 No-deletion rule

The coverage audit now treats the integrated front manuscript and ancillary data as the submission record. Duplicated source-PDF appendices are removed from the arXiv source package so that the paper does not cite or repackage earlier author manuscripts as external works.

Chapter 17

Expanded Defect Calculus proof capsules

17.1 Fixed-point and active-graph constraints

Proposition 17.1 (Fixed-point constraints). *A configuration x is a fixed point of the base map if and only if every active site has $5 \leq N_i(x) \leq 9$ and every inactive site avoids the birth counts $N_i(x) \in \{5, 6, 7\}$.*

Proof. A fixed active site must satisfy the survival rule; a fixed inactive site must fail the birth rule. These two local conditions are independent at each site and exactly characterize $P(x) = x$. $\square$

Proposition 17.2 (Active-degree bounds). *If x is a nonzero fixed point and $G_A(x)$ is the graph on active sites with Moore-neighborhood edges, then every active vertex has degree between 5 and 9, and therefore*

$$5 \leq \frac{2|E(G_A)|}{|A|} \leq 9.$$

Proof. For an active site, $N_i(x)$ is exactly its active Moore degree. The fixed-point survival condition gives $5 \leq N_i \leq 9$ at every active vertex. Summing all active degrees gives $2|E(G_A)|$ and yields the average-degree bound. $\square$

17.2 Slab rejection and dimensional selectivity

Proposition 17.3 (Filled slabs are rejected). *A union of at least two consecutive full coordinate sheets dies in one step under the base rule.*

Proof. Each active site in a slab of thickness at least two sees eight active neighbors in its own sheet and at least nine in an adjacent sheet, so it has at least 17 active neighbors and exceeds the survival ceiling 9. Inactive sites outside the slab see counts 0, 9, or 18, none of which lies in the inactive birth window $\{5, 6, 7\}$. $\square$

Remark 17.4 (Why this matters). *The rule is not a generic growth rule. It rejects dust, lines, and bulk while preserving one-cell surfaces. This is the first reason the program is organized around surface defects rather than volume filling.*

17.3 The planar selector in channel form

Let $D \subset T_L$ be a missing-cell mask in a stable sheet. For $u \in T_L$, recall

$$q_D(u) = \sum_{\eta \in \{-1, 0, 1\}^2 \setminus \{0\}} \mathbf{1}\{u + \eta \in D\}, \quad r_D(u) = \sum_{\eta \in \{-1, 0, 1\}^2} \mathbf{1}\{u + \eta \in D\}.$$

The source theorem states that active sheet cells survive exactly when $q_D \leq 3$, in-plane missing cells are born exactly when $1 \leq q_D \leq 3$, and normal-layer cells are born exactly when $2 \leq r_D \leq 4$.

Proof capsule. On the sheet plane, a site sees only the eight planar neighbors of the original sheet, minus the missing sites in its open footprint. Hence the raw count is $8 - q_D(u)$. If the site is active, the survival rule accepts $5 \leq 8 - q_D \leq 9$, equivalent to $q_D \leq 3$. If the site is a missing in-plane defect, the birth rule accepts $5 \leq 8 - q_D \leq 7$, equivalent to $1 \leq q_D \leq 3$. A cell one normal layer away sees the closed 3×3 footprint below or above it, so the raw count is $9 - r_D(u)$; the birth rule accepts $5 \leq 9 - r_D \leq 7$, equivalent to $2 \leq r_D \leq 4$. All layers two or more steps away see no active neighbors on the first update. $\square$

17.4 Stable pore gas

Corollary 17.5 (Stable isolated pore gas). *If every closed 3×3 footprint contains at most one missing sheet site, then the defective sheet is a fixed point.*

Proof. The condition gives $r_D \leq 1$ everywhere, so no normal-layer birth can satisfy $2 \leq r_D \leq 4$. Each missing pore has $q_D = 0$, so it is not reborn in-plane. Every active sheet site has $q_D \leq 1 \leq 3$, so all active cells survive. $\square$

17.5 Cracks, line sources, and line-hole edge cases

The source paper separates isolated pores from slits, cracks, and line holes. The front-manuscript summary is:

1. isolated pores freeze because they never reach the normal-layer birth window;
2. adjacent missing cells create locations with $r_D \in \{2, 3, 4\}$ and therefore emit normal-layer births;
3. long line holes are edge cases whose first-step response is controlled by endpoint and edge footprints rather than area alone.

These cases remain theorem-level when stated as first-step selector consequences. Multi-step relaxation beyond the first response remains empirical-atlas material unless restricted to a proven subfamily.

Chapter 18

Rectangle, channel, prime, work, and reconstruction proof capsules

18.1 Rectangle zero-death subfamily

For an $a \times b$ rectangular puncture with $a, b \geq 2$ in an otherwise stable sheet, the first response is a closed zero-death subfamily. Its raw first-step normal birth law is

$$B_0(a, b) = 4a + 4b + 12 = 2P + 12,$$

where $P = 2a + 2b$ is the rectangle perimeter in sheet units.

Proof capsule. The active-sheet death test is $q_D \geq 4$. Around a filled rectangle, active cells adjacent to the rectangle boundary see at most three missing cells in their open planar footprint, so they survive. Normal-layer candidates are born exactly where the closed footprint contains 2, 3, or 4 rectangle cells. Counting the two normal layers above and below the rectangle gives fixed corner, edge, and near-corner contributions that sum to $4a + 4b + 12$. $\square$

18.2 Rectangle channel spectrum

The rectangle first-step births split into the raw channel spectrum

$$(N_5, N_6, N_7) = (12, 4a + 4b - 16, 16).$$

Equivalently, the B_5 group is fixed at 12, the B_7 group is fixed at 16, and the edge-extensive part lives in B_6 .

Proof capsule. For a normal-layer candidate, the raw count is $9 - r_D$. Thus B_7 corresponds to $r_D = 2$, B_6 to $r_D = 3$, and B_5 to $r_D = 4$. The closed footprints at the rectangle corners, edges, and corner-adjacent positions can be counted separately. The $r_D = 4$ corner groups give the fixed 12 count; the $r_D = 2$ shallow edge groups give the fixed 16 count; the remaining edge-length contribution gives $4a + 4b - 16$ in channel B_6 . $\square$

18.3 Prime-corrected rectangle model

The default prime-sharpened model removes the raw B_7 birth group from the rectangle response, leaving

$$B_\varphi(a, b) = B_0(a, b) - 16 = 4a + 4b - 4.$$

This is a model-tagged response, not the raw base rule. It is useful as a sharpened diagnostic because B_7 is the high birth edge and often behaves like a plasticity gate.

18.4 First-step work laws

Given nonnegative channel weights I_5, I_6, I_7 , the first-step rectangle birth work is

$$W_B(a, b) = 12I_5 + (4a + 4b - 16)I_6 + 16I_7.$$

For the prime-sharpened response that removes B_7 ,

$$W_{B,\varphi}(a, b) = 12I_5 + (4a + 4b - 16)I_6.$$

These are exact once the channel spectrum and weights are fixed. The weights themselves are bookkeeping choices unless a separate physical cost model is supplied.

18.5 Rectangle reconstruction boundary

Under the rectangle prior, the pair of channel counts is sufficient to recover the rectangle perimeter and therefore the family of possible side pairs. The boundary is limited: the channel spectrum reconstructs a rectangle class under a rectangle assumption; it does not reconstruct arbitrary masks.

Proof capsule. For rectangles, $N_6 = 4a + 4b - 16$ and $B_0 = 4a + 4b + 12$. Either value recovers $a + b$. The possible side pairs are therefore the integer pairs (a, b) with the recovered sum and the imposed side constraints. Without the rectangle prior, many masks can share the same channel counts, so the result is not arbitrary shape reconstruction. $\square$

Chapter 19

PooleQ/P proof capsules

19.1 Channel disjointness

For each site exactly one value of N_i holds. Therefore the birth channels $B_k = (1 - x_i)\mathbf{1}\{N_i = k\}$ and survival channels $S_k = x_i\mathbf{1}\{N_i = k\}$ are disjoint. The raw Poole update is the collapsed sum

$$P(x)_i = \sum_{k=5}^7 B_k(i; x) + \sum_{k=5}^9 S_k(i; x).$$

The Q/P move is to preserve the vector of channels before collapse.

19.2 Expected p-bit channels

Let $X_j \sim \text{Bernoulli}(p_j)$ independently and let

$$G_i(z) = \prod_{j \in N_{26}(i)} (1 - p_j + p_j z).$$

Then $\Pr[N_i(X) = k] = [z^k]G_i(z)$ and

$$E[B_k(i; X)] = (1 - p_i)[z^k]G_i(z), \quad E[S_k(i; X)] = p_i[z^k]G_i(z).$$

Consequently

$$q_i = E[P(X)_i] = C_i^{5:7} + p_i C_i^{8:9}.$$

This identity is the probabilistic form of local hysteresis: the center probability weights only the upper memory band.

19.3 Differentiability capsule

For a neighbor $\ell \in N_{26}(i)$,

$$\frac{\partial}{\partial p_\ell} [z^k]G_i(z) = [z^k] \left((z - 1) \prod_{j \in N_{26}(i), j \neq \ell} (1 - p_j + p_j z) \right).$$

For the center site,

$$\frac{\partial q_i}{\partial p_i} = C_i^{8:9}.$$

This gives a differentiable probability-layer version of the channel transform under the independent local product approximation.

19.4 Correlation residue and Poole coherence

Given repeated measured samples, empirical channel rates can be compared against the independent prediction formed from observed local marginals. The channel-wise residue vector is

$$R_i = (\hat{B}_5 - E_{ind}B_5, \hat{B}_6 - E_{ind}B_6, \hat{B}_7 - E_{ind}B_7, \hat{S}_8 - E_{ind}S_8, \hat{S}_9 - E_{ind}S_9)_i,$$

and a global local-correlation statistic is

$$C_P = \sum_i \|R_i\|_2^2.$$

Large C_P indicates that product marginals fail to explain local channel rates. It is not by itself evidence for a particular physical noise model.

19.5 Exact-cardinality defect logic

For a defective stable sheet and a normal-layer candidate, the active neighbor count is $N = 9 - r_D$. Hence

$$B_7 \Leftrightarrow r_D = 2, \quad B_6 \Leftrightarrow r_D = 3, \quad B_5 \Leftrightarrow r_D = 4.$$

With selected input defects and fixed bias defects in the same 3×3 footprint, this becomes exact-cardinality detection. If s input defects and b bias defects give $r = s + b$, then the birth channels select the exact values $s = 2 - b$, $s = 3 - b$, and $s = 4 - b$.

19.6 Gate and adder boundary

The source Q/P paper shows how biased defect footprints can implement raw Boolean gate readouts and channel-resolved adder readouts. The boundary is important: these are exact for channel-resolved readout, spatially demultiplexed layouts, or instrumented local feature readout. A one-state uninstrumented cellular automaton that observes only $P(x)_i$ collapses the channel identity.

19.7 Syndrome-field boundary

For qubit readout strings or stabilizer syndrome histories, PooleQ/P is a classical transform applied after measurement. It does not copy, clone, or reconstruct an unknown quantum state. Its role is local geometry spectrometry: mapping already-measured binary fields to interpretable channel signatures.

Chapter 20

PooleQ/P verifier and benchmark import v1.2

This chapter imports the PooleQ/P verifier and benchmark layer into the front manuscript. The purpose is not to elevate every Q/P application into a theorem. The purpose is to separate three claim types cleanly: exact footprint identities, finite same-density shuffled-control benchmarks, and bounded spectrometer interpretation.

20.1 v1.2 Q/P pass/fail summary

The exact footprint checks are first-update local statements under the sheet-footprint assumptions already stated above. They check the raw biased-footprint Boolean primitives, one-input identity/not cases, half-adder channel readout, full-adder channel readout, and the cardinality-to-channel equivalence.

Table 20.1: v1.2 Q/P footprint verifier pass/fail summary.

Check	Matches	Cases
Raw two-input gates: AND, OR, NAND, NOR	16	16
One-input identity/not	4	4
Half-adder channel readout	4	4
Full-adder channel readout	8	8
Cardinality-to-channel table	10	10

20.2 Exact-cardinality and raw gate verifier

For a normal-layer candidate adjacent to a defective sheet, r missing footprint cells give $N = 9 - r$ active neighbors. The raw collapsed birth event is therefore

$$Y = \mathbf{1}\{2 \leq r \leq 4\}.$$

With two inputs A, B and a fixed bias count c , the verifier checks

$$Y_{2,c}(A, B) = \mathbf{1}\{2 \leq c + A + B \leq 4\}.$$

The four bias choices reproduce the expected raw gates.

Table 20.2: Raw biased-footprint gate truth table.

Gate	A	B	bias c	Y	expected
AND	0	0	0	0	0
AND	0	1	0	0	0
AND	1	0	0	0	0
AND	1	1	0	1	1
OR	0	0	1	0	0
OR	0	1	1	1	1
OR	1	0	1	1	1
OR	1	1	1	1	1
NAND	0	0	3	1	1
NAND	0	1	3	1	1
NAND	1	0	3	1	1
NAND	1	1	3	0	0
NOR	0	0	4	1	1
NOR	0	1	4	0	0
NOR	1	0	4	0	0
NOR	1	1	4	0	0

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The surrounding caption, tables, and ancillary data contain the reported result.

Figure 20.1: v1.2 Q/P footprint verifier pass fractions. These are exact finite truth-table checks, not statistical benchmarks.

20.3 Channel-resolved adder verifier

For the half-adder, the footprint uses one bias defect so that $r = A + B + 1$. Then B_7 is the exact-one channel and B_6 is the exact-two channel:

$$\text{SUM} = B_7, \quad \text{CARRY} = B_6.$$

For the full-adder with inputs A, B, C_{in} and the same one-bias layout,

$$B_7 = \mathbf{1}\{s = 1\}, \quad B_6 = \mathbf{1}\{s = 2\}, \quad B_5 = \mathbf{1}\{s = 3\},$$

where $s = A + B + C_{in}$. Therefore

$$\text{SUM} = B_7 \vee B_5, \quad \text{CARRY} = B_6 \vee B_5.$$

Table 20.3: Full-adder channel verifier truth table.

A	B	C_{in}	s	B_7	B_6	B_5	SUM	CARRY
0	0	0	0	0	0	0	0	0
0	0	1	1	1	0	0	1	0
0	1	0	1	1	0	0	1	0
0	1	1	2	0	1	0	0	1
1	0	0	1	1	0	0	1	0
1	0	1	2	0	1	0	0	1
1	1	0	2	0	1	0	0	1
1	1	1	3	0	0	1	1	1

Remark 20.1 (Adder boundary). *The full-adder statement is channel-resolved. If the experiment records only the collapsed raw event $Y = B_5 \vee B_6 \vee B_7$, the three-input layout collapses to $A \vee B \vee C_{in}$ for the one-bias case. The adder therefore requires channel readout or spatial demultiplexing.*

20.4 Same-density shuffled-control benchmark import

The Q/P benchmark layer is a finite measurement-side result. It asks whether channel spectra distinguish structured fields from same-density nulls after active count has been controlled. In the v5.5 decomposed benchmark, all listed fields were active-density matched and exact same-density shuffled controls were included.

Table 20.4: v5.5 decomposed robust capture scores imported into the front manuscript.

Class	R_B	R_H	R_ψ	R_C	Signature
IID null	0.21	0.02	0.16	0.28	null-like
Straight lines	3.49	2.19	3.28	5.28	mixed geometry
Lines shuffled	0.24	0.03	0.17	0.31	null-like
Random-walk chains	3.62	5.63	4.15	7.88	high-support-led
Walks shuffled	0.23	0.02	0.19	0.31	null-like
Branching chains	3.87	3.66	3.88	6.61	mixed geometry
Branches shuffled	0.21	0.02	0.14	0.27	null-like
Compact bursts	3.12	7.06	5.00	9.19	high-support-led
Bursts shuffled	0.23	0.02	0.16	0.29	null-like
Sheets	3.50	7.40	5.31	9.76	high-support-led
Sheets shuffled	0.22	0.04	0.16	0.29	null-like

Table 20.5: Paired same-density shuffled controls from the v5.5 benchmark.

Structured class	R_C	Shuffled R_C	Drop	Signature
Straight lines	5.28	0.31	4.97	mixed geometry
Random-walk chains	7.88	0.31	7.57	high-support-led
Branching chains	6.61	0.27	6.34	mixed geometry
Compact bursts	9.19	0.29	8.90	high-support-led
Sheets	9.76	0.29	9.47	high-support-led

The preserved v5.6 spectrometer shares express the direction of the signal after null separation. They are not interpreted for null-like classes.

Table 20.6: v5.6 geometry-spectrometer shares imported from the decomposed benchmark.

Class	R_C	Birth share	High-support share	Strain share	Readout
IID null	0.28	0.64	0.01	0.36	null-like
Straight lines	5.28	0.44	0.17	0.39	mixed geometry
Random-walk chains	7.88	0.21	0.51	0.28	high-support-led
Branching chains	6.61	0.34	0.31	0.35	mixed geometry
Compact bursts	9.19	0.11	0.59	0.30	high-support-led
Sheets	9.76	0.13	0.58	0.30	high-support-led

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The surrounding caption, tables, and ancillary data contain the reported result.

Figure 20.2: Combined non-redundant robust score R_C for the v5.5 Q/P benchmark. Structured fields remain high while IID and same-density shuffled controls remain null-like.

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The surrounding caption, tables, and ancillary data contain the reported result.

Figure 20.3: Decomposed robust components for the v5.5 Q/P benchmark. The decomposition gives an interpretable geometry signature rather than only a scalar anomaly score.

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The surrounding caption, tables, and ancillary data contain the reported result.

Figure 20.4: v5.6 geometry-spectrometer share interpretation for the non-null structured fields.

20.5 v1.2 Q/P benchmark boundary

The Q/P benchmark supports the following bounded claim: PooleQ/P maps already-measured binary fields into local channel spectra that separate structured fields from exact same-density shuffled controls in the tested synthetic families. It does not establish state reconstruction, quantum amplitude capture, or an autonomous computing architecture. The exact footprint identities are theorem/verifier-level; the shuffled-control separation is benchmark-level; and the geometry-spectrometer interpretation is a bounded reading of the decomposed scores.

Chapter 21

Signed-membrane benchmark tables imported into the front manuscript

The following tables are imported directly from the recovered original signed reaction-membrane benchmark bundle. The tables use the manuscript-safe interpretation of M_P and J : they are benchmark descriptors, not physical energy or thrust laws.

Table 21.1: Signed-membrane induction table imported from the full benchmark bundle.

condition	maintainer	init/ticks	J	Q_ρ	σ_y	$\epsilon_\pm$	good
mirrored0_hybrid	hybrid_guard	mirrored/0	0.700	0.507	0.080	0.007	100.0%
mirrored10_hybrid	hybrid_guard	mirrored/10	0.745	0.520	0.007	0.004	100.0%
mirrored10_membrane	passive_lock	mirrored/10	0.785	0.548	0.000	0.003	100.0%
mirrored20_hybrid	hybrid_guard	mirrored/20	0.751	0.522	0.001	0.003	100.0%
preformed_sheet_hybrid	hybrid_guard	preformed_sheet/0	0.254	0.406	1.713	0.164	43.8%
random_hybrid	hybrid_guard	random/0	0.492	0.450	0.648	0.072	68.8%

Table 21.2: Signed-membrane maintenance table imported into the front manuscript.

condition	maintainer	J	Q_ρ	σ_y	$\epsilon_\pm$	moves
maintain_focus	focus_only	0.744	0.519	0.007	0.004	4.163
maintain_hybrid	hybrid_guard	0.745	0.520	0.007	0.004	4.158
maintain_interfaceblind	interface.blind	0.761	0.529	0.000	0.003	1.613
maintain_membrane	passive_lock	0.785	0.548	0.000	0.003	0.000
maintain_random	same_move_random	0.771	0.536	0.006	0.003	1.554
maintain_static	broad_static_forcing	0.602	0.485	0.191	0.014	168.467

Table 21.3: Perturbation-stress results imported from the signed benchmark bundle.

perturb.	controller	J	Q_ρ	σ_y	good
0.0	focus_only	0.746	0.522	0.001	1.000
0.0	hybrid_guard	0.746	0.522	0.001	1.000
0.0	passive_lock	0.781	0.552	0.000	1.000
0.0	broad_static_forcing	0.609	0.485	0.177	1.000
0.01	focus_only	0.736	0.520	0.019	1.000
0.01	hybrid_guard	0.735	0.520	0.018	1.000
0.01	passive_lock	0.772	0.545	0.000	1.000
0.01	broad_static_forcing	0.616	0.485	0.152	1.000
0.03	focus_only	0.736	0.518	0.007	1.000
0.03	hybrid_guard	0.736	0.518	0.007	1.000
0.03	passive_lock	0.759	0.536	0.006	1.000
0.03	broad_static_forcing	0.601	0.484	0.205	1.000

perturb.	controller	J	Q_ρ	σ_y	good
0.05	focus_only	0.726	0.514	0.026	1.000
0.05	hybrid_guard	0.725	0.513	0.025	1.000
0.05	passive_lock	0.745	0.533	0.028	1.000
0.05	broad_static_forcing	0.605	0.484	0.186	1.000
0.1	focus_only	0.711	0.515	0.062	1.000
0.1	hybrid_guard	0.709	0.516	0.072	1.000
0.1	passive_lock	0.725	0.526	0.047	1.000
0.1	broad_static_forcing	0.613	0.484	0.167	1.000

rule	gentle J	good
B5-7.S4-11_k3	0.756	100.0%
B5-7.S3-12_k3	0.750	100.0%
B5-7.S3-11_k3	0.748	100.0%
B5-7.S5-12_k3	0.742	100.0%
B5-7.S5-10_k3	0.651	100.0%
B5-7.S5-9_k3	0.560	100.0%

Table 21.4: Signed-membrane survival-window ridge table.

Table 21.5: Complete condensed geometry and boundary robustness table.

L	boundary	controller	J	Q_ρ	σ_y	$\epsilon_\pm$	good
32	absorbing_outlet	focus_only	0.830	0.616	0.000	0.005	100.0%
32	absorbing_outlet	hybrid_guard	0.830	0.616	0.000	0.005	100.0%
32	absorbing_outlet	interface_blind	0.837	0.616	0.000	0.005	100.0%
32	absorbing_outlet	passive_lock	0.872	0.648	0.000	0.003	100.0%
32	absorbing_outlet	broad_static_forcing	0.760	0.591	0.110	0.006	100.0%
32	zero	focus_only	0.895	0.644	0.000	0.005	100.0%
32	zero	hybrid_guard	0.895	0.644	0.000	0.005	100.0%
32	zero	interface_blind	0.906	0.645	0.000	0.005	100.0%
32	zero	passive_lock	0.944	0.682	0.000	0.003	100.0%
32	zero	broad_static_forcing	0.823	0.624	0.131	0.008	100.0%
40	absorbing_outlet	focus_only	0.701	0.498	0.000	0.003	100.0%
40	absorbing_outlet	hybrid_guard	0.701	0.498	0.000	0.003	100.0%
40	absorbing_outlet	interface_blind	0.712	0.510	0.000	0.003	100.0%
40	absorbing_outlet	passive_lock	0.734	0.526	0.000	0.003	100.0%
40	absorbing_outlet	broad_static_forcing	0.573	0.463	0.174	0.008	100.0%
40	zero	focus_only	0.743	0.518	0.000	0.003	100.0%
40	zero	hybrid_guard	0.743	0.518	0.000	0.003	100.0%
40	zero	interface_blind	0.760	0.531	0.000	0.003	100.0%
40	zero	passive_lock	0.779	0.547	0.000	0.002	100.0%
40	zero	broad_static_forcing	0.603	0.480	0.197	0.010	100.0%
48	absorbing_outlet	focus_only	0.626	0.420	0.021	0.003	75.0%
48	absorbing_outlet	hybrid_guard	0.627	0.420	0.021	0.003	75.0%
48	absorbing_outlet	interface_blind	0.646	0.432	0.000	0.003	87.5%
48	absorbing_outlet	passive_lock	0.656	0.442	0.000	0.003	75.0%
48	absorbing_outlet	broad_static_forcing	0.589	0.410	0.116	0.004	37.5%
48	zero	focus_only	0.657	0.434	0.026	0.004	100.0%
48	zero	hybrid_guard	0.657	0.434	0.025	0.004	100.0%
48	zero	interface_blind	0.678	0.447	0.000	0.003	100.0%
48	zero	passive_lock	0.690	0.458	0.000	0.003	100.0%
48	zero	broad_static_forcing	0.616	0.424	0.136	0.005	62.5%

Table 21.6: Complete condensed mirror-quality basin table.

mirror q	controller	J	Q_ρ	σ_y	$\epsilon_\pm$	good
0.25	hybrid_guard	0.196	0.504	3.123	0.207	18.8%
0.25	passive_lock	0.200	0.506	3.146	0.205	0.0%

mirror q	controller	J	Q_ρ	σ_y	$\epsilon_\pm$	good
0.5	hybrid_guard	0.405	0.440	1.399	0.088	43.8%
0.5	passive_lock	0.414	0.450	1.425	0.088	6.2%
0.75	hybrid_guard	0.552	0.464	0.472	0.035	87.5%
0.75	passive_lock	0.556	0.472	0.486	0.035	62.5%
0.9	hybrid_guard	0.622	0.485	0.295	0.021	100.0%
0.9	passive_lock	0.633	0.492	0.280	0.019	100.0%
1.0	hybrid_guard	0.745	0.520	0.007	0.004	100.0%
1.0	passive_lock	0.785	0.548	0.000	0.003	100.0%
1.0	hybrid_guard	0.492	0.450	0.648	0.072	68.8%
1.0	passive_lock	0.490	0.449	0.645	0.075	31.2%

Chapter 22

Signed-membrane benchmark deep import v1.3

This v1.3 pass moves the recovered signed reaction-membrane benchmark bundle further into the front manuscript. The previous front manuscript already imported the paper-facing induction, maintenance, stress, rule-ridge, geometry, and mirror-quality tables. This chapter adds the certification layer: file manifest, seed counts, certificate flags, paired seed comparisons, original figures, and generated pairwise-lift summaries.

The benchmark remains a finite cellular-automaton benchmark. Matter–antimatter language is used only as motivating terminology: modern antimatter work relies on specialized accelerator, deceleration, trapping, confinement, and transport infrastructure, and propulsion-oriented references describe major production, storage, and utilization barriers [8, 9, 10, 11, 12]. The present manuscript does not claim antimatter production, storage, shielding, propulsion, or a physical energy law. The quantities M_P , Q_ρ , σ_y , $\epsilon_\pm$, and J are morphology and scoring descriptors for signed reaction-interface basins.

22.1 Recovered benchmark bundle and certificate

The recovered benchmark bundle contains by-seed CSVs, by-setting CSVs, paired-comparison CSVs, paper-facing tables, figures, a configuration file, a certificate, and a summary file. The file manifest is included to make the data path explicit.

Table 22.1: Signed-membrane recovered benchmark file dimensions for v1.3.

file	rows	columns	unique conditions	unique seeds
benchmark_all_by_seed.csv	1284	87	109	48
benchmark_induction_by_seed.csv	96	87	6	16
benchmark_maintenance_by_seed.csv	96	87	6	16
benchmark_stress_by_seed.csv	300	87	25	12
benchmark_rule_ridge_by_seed.csv	360	87	30	12
benchmark_geometry_by_seed.csv	240	87	30	8
benchmark_mirror_quality_by_seed.csv	192	87	12	16
benchmark_induction_by_setting.csv	6	119	6	
benchmark_maintenance_by_setting.csv	6	119	6	
benchmark_stress_by_setting.csv	25	119	25	
benchmark_rule_ridge_by_setting.csv	30	119	30	
benchmark_geometry_by_setting.csv	30	119	30	
benchmark_mirror_quality_by_setting.csv	12	119	12	

The certificate records the benchmark seed counts and the main pass/fail conclusions used by the signed-membrane section. The safest reading is that the certificate validates the benchmark families under their tested parameters, not an unlimited rule-space or geometry-space theorem.

Table 22.2: Signed-membrane benchmark certificate imported into the front manuscript.

certificate field	value
version	benchmark_v1
benchmark_name	Poole Signed Reaction Membrane Benchmark v1
final_reference_rule	B5-7_S3-11_k3
best_rule.in_benchmark_rule_ridge	B5-7_S4-11_k3
seed_count_main	16
seed_count_ridge	12
seed_count_geometry	8
seed_count_stress	12
induction_practical_certified_mirrored10_hybrid_vs_random	True
induction_good_regime_certified_mirrored10_hybrid_vs_random	True
stress_membrane_beats_static_all_perturbations	True
geometry_membrane_beats_static_all_L_boundary	True
best_rule_beats_canonical_all_gentle_controllers	True
perfect_mirror_practical_certified_vs_random_membrane	True
mirror90_practical_certified_vs_random_membrane	True

22.2 Pairwise seed certification

The paired CSVs compare candidate and control conditions seed-by-seed. This is stronger than comparing only unpaired means, because the same seed structure is used to estimate paired differences and confidence intervals.

Table 22.3: Condensed signed-membrane paired practical-score certification by evidence family.

family	comps.	cert.	mean diff	min diff	max diff	mean win
geometry	12	12	0.078	0.029	0.177	1.000
induction	5	5	0.268	0.045	0.490	0.950
maintenance	5	3	0.065	-0.026	0.183	0.613
mirror_quality	8	8	0.226	0.122	0.371	0.984
rule_ridge	25	23	0.151	0.027	0.214	0.953
stress	15	15	0.103	0.016	0.172	0.994

The complete paired practical-score table is included below. Rows with positive certified status support the signed-membrane claim that prepared membranes are induced and maintained by local geometry and gentle dynamics, while broad forcing and random controls do not generate the same basin.

Table 22.4: Complete v1.3 paired practical-score table from the signed benchmark. Each row compares a candidate condition against a matched control using paired seeds.

group	comparison	candidate	control	n	cand.	ctrl.	diff	win
induction	induction_vs_random	mirrored10_hybrid	random_hybrid	16	0.745	0.492	0.252	1.000
induction	induction_vs_random	mirrored20_hybrid	random_hybrid	16	0.751	0.492	0.259	1.000
induction	warmup_gain	mirrored10_hybrid	mirrored0_hybrid	16	0.745	0.700	0.045	0.750
induction	passive_induction_vs_random	mirrored10_membrane	random_hybrid	16	0.785	0.492	0.293	1.000
induction	induction_vs_forced_sheet	mirrored10_hybrid	preformed_sheet_hybrid	16	0.745	0.254	0.490	1.000
maintenance	passive_vs_static	maintain_membrane	maintain_static	16	0.785	0.602	0.183	1.000
maintenance	passive_vs_hybrid	maintain_membrane	maintain_hybrid	16	0.785	0.745	0.040	1.000
maintenance	hybrid_vs_random	maintain_hybrid	maintain_random	16	0.745	0.771	-0.026	0.062
maintenance	hybrid_vs_interfaceblind	maintain_hybrid	maintain_interfaceblind	16	0.745	0.761	-0.017	0.000
maintenance	focus_vs_static	maintain_focus	maintain_static	16	0.744	0.602	0.143	1.000
stress	membrane_vs_static_stress	stress_p00_membrane	stress_p00_static	12	0.781	0.609	0.172	1.000
stress	membrane_vs_hybrid_stress	stress_p00_membrane	stress_p00_hybrid	12	0.781	0.746	0.035	1.000
stress	interfaceblind_vs_static_stress	stress_p00_interfaceblind	stress_p00_static	12	0.763	0.609	0.154	1.000
stress	membrane_vs_static_stress	stress_p01_membrane	stress_p01_static	12	0.772	0.616	0.156	1.000
stress	membrane_vs_hybrid_stress	stress_p01_membrane	stress_p01_hybrid	12	0.772	0.735	0.037	1.000
stress	interfaceblind_vs_static_stress	stress_p01_interfaceblind	stress_p01_static	12	0.756	0.616	0.139	1.000
stress	membrane_vs_static_stress	stress_p03_membrane	stress_p03_static	12	0.759	0.601	0.158	1.000
stress	membrane_vs_hybrid_stress	stress_p03_membrane	stress_p03_hybrid	12	0.759	0.736	0.022	1.000
stress	interfaceblind_vs_static_stress	stress_p03_interfaceblind	stress_p03_static	12	0.745	0.601	0.144	1.000
stress	membrane_vs_static_stress	stress_p05_membrane	stress_p05_static	12	0.745	0.605	0.140	1.000
stress	membrane_vs_hybrid_stress	stress_p05_membrane	stress_p05_hybrid	12	0.745	0.725	0.020	1.000
stress	interfaceblind_vs_static_stress	stress_p05_interfaceblind	stress_p05_static	12	0.733	0.605	0.128	1.000
stress	membrane_vs_static_stress	stress_p10_membrane	stress_p10_static	12	0.725	0.613	0.112	1.000
stress	membrane_vs_hybrid_stress	stress_p10_membrane	stress_p10_hybrid	12	0.725	0.709	0.016	0.917
stress	interfaceblind_vs_static_stress	stress_p10_interfaceblind	stress_p10_static	12	0.726	0.613	0.113	1.000
rule_ridge	rule_vs_canonical	rule_B57_S510_k3_membrane	rule_B57_S59_k3_membrane	12	0.717	0.564	0.153	1.000
rule_ridge	rule_vs_canonical	rule_B57_S510_k3_hybrid	rule_B57_S59_k3_hybrid	12	0.623	0.558	0.065	0.833
rule_ridge	rule_vs_canonical	rule_B57_S510_k3_focus	rule_B57_S59_k3_focus	12	0.619	0.558	0.062	1.000
rule_ridge	rule_vs_canonical	rule_B57_S510_k3_interfaceblind	rule_B57_S59_k3_interfaceblind	12	0.647	0.562	0.085	0.917
rule_ridge	rule_vs_canonical	rule_B57_S510_k3_static	rule_B57_S59_k3_static	12	0.512	0.485	0.027	0.500
rule_ridge	rule_vs_canonical	rule_B57_S411_k3_membrane	rule_B57_S59_k3_membrane	12	0.777	0.564	0.213	1.000
rule_ridge	rule_vs_canonical	rule_B57_S411_k3_hybrid	rule_B57_S59_k3_hybrid	12	0.744	0.558	0.186	1.000
rule_ridge	rule_vs_canonical	rule_B57_S411_k3_focus	rule_B57_S59_k3_focus	12	0.744	0.558	0.186	1.000
rule_ridge	rule_vs_canonical	rule_B57_S411_k3_interfaceblind	rule_B57_S59_k3_interfaceblind	12	0.758	0.562	0.196	1.000
rule_ridge	rule_vs_canonical	rule_B57_S411_k3_static	rule_B57_S59_k3_static	12	0.561	0.485	0.077	0.917
rule_ridge	rule_vs_canonical	rule_B57_S512_k3_membrane	rule_B57_S59_k3_membrane	12	0.771	0.564	0.207	1.000
rule_ridge	rule_vs_canonical	rule_B57_S512_k3_hybrid	rule_B57_S59_k3_hybrid	12	0.724	0.558	0.167	1.000
rule_ridge	rule_vs_canonical	rule_B57_S512_k3_focus	rule_B57_S59_k3_focus	12	0.724	0.558	0.167	1.000
rule_ridge	rule_vs_canonical	rule_B57_S512_k3_interfaceblind	rule_B57_S59_k3_interfaceblind	12	0.748	0.562	0.186	1.000
rule_ridge	rule_vs_canonical	rule_B57_S512_k3_static	rule_B57_S59_k3_static	12	0.511	0.485	0.027	0.667
rule_ridge	rule_vs_canonical	rule_B57_S311_k3_membrane	rule_B57_S59_k3_membrane	12	0.771	0.564	0.207	1.000
rule_ridge	rule_vs_canonical	rule_B57_S311_k3_hybrid	rule_B57_S59_k3_hybrid	12	0.734	0.558	0.176	1.000
rule_ridge	rule_vs_canonical	rule_B57_S311_k3_focus	rule_B57_S59_k3_focus	12	0.734	0.558	0.176	1.000
rule_ridge	rule_vs_canonical	rule_B57_S311_k3_interfaceblind	rule_B57_S59_k3_interfaceblind	12	0.753	0.562	0.191	1.000
rule_ridge	rule_vs_canonical	rule_B57_S311_k3_static	rule_B57_S59_k3_static	12	0.612	0.485	0.127	1.000
rule_ridge	rule_vs_canonical	rule_B57_S312_k3_membrane	rule_B57_S59_k3_membrane	12	0.778	0.564	0.214	1.000
rule_ridge	rule_vs_canonical	rule_B57_S312_k3_hybrid	rule_B57_S59_k3_hybrid	12	0.734	0.558	0.176	1.000
rule_ridge	rule_vs_canonical	rule_B57_S312_k3_focus	rule_B57_S59_k3_focus	12	0.734	0.558	0.176	1.000
rule_ridge	rule_vs_canonical	rule_B57_S312_k3_interfaceblind	rule_B57_S59_k3_interfaceblind	12	0.755	0.562	0.193	1.000
rule_ridge	rule_vs_canonical	rule_B57_S312_k3_static	rule_B57_S59_k3_static	12	0.613	0.485	0.129	1.000
geometry	membrane_vs_static_geometry	geom.L32_zero_membrane	geom.L32_zero_static	8	0.944	0.823	0.122	1.000
geometry	membrane_vs_hybrid_geometry	geom.L32_zero_membrane	geom.L32_zero_hybrid	8	0.944	0.895	0.049	1.000
geometry	membrane_vs_static_geometry	geom.L32_absorbing_outlet_membrane	geom.L32_absorbing_outlet_static	8	0.872	0.760	0.111	1.000
geometry	membrane_vs_hybrid_geometry	geom.L32_absorbing_outlet_membrane	geom.L32_absorbing_outlet_hybrid	8	0.872	0.830	0.042	1.000
geometry	membrane_vs_static_geometry	geom.L40_zero_membrane	geom.L40_zero_static	8	0.779	0.603	0.177	1.000
geometry	membrane_vs_hybrid_geometry	geom.L40_zero_membrane	geom.L40_zero_hybrid	8	0.779	0.743	0.036	1.000
geometry	membrane_vs_static_geometry	geom.L40_absorbing_outlet_membrane	geom.L40_absorbing_outlet_static	8	0.734	0.573	0.162	1.000
geometry	membrane_vs_hybrid_geometry	geom.L40_absorbing_outlet_membrane	geom.L40_absorbing_outlet_hybrid	8	0.734	0.701	0.033	1.000
geometry	membrane_vs_static_geometry	geom.L48_zero_membrane	geom.L48_zero_static	8	0.690	0.616	0.074	1.000
geometry	membrane_vs_hybrid_geometry	geom.L48_zero_membrane	geom.L48_zero_hybrid	8	0.690	0.657	0.033	1.000
geometry	membrane_vs_static_geometry	geom.L48_absorbing_outlet_membrane	geom.L48_absorbing_outlet_static	8	0.656	0.589	0.066	1.000
geometry	membrane_vs_hybrid_geometry	geom.L48_absorbing_outlet_membrane	geom.L48_absorbing_outlet_hybrid	8	0.656	0.627	0.029	1.000
mirror_quality	perfect_mirror_vs_random	mirror.q100_membrane	mirror_random_membrane	16	0.785	0.490	0.295	1.000
mirror_quality	90pct_mirror_vs_random	mirror.q090_membrane	mirror_random_membrane	16	0.633	0.490	0.142	0.938
mirror_quality	perfect_vs_90pct	mirror.q100_membrane	mirror.q090_membrane	16	0.785	0.633	0.153	1.000

group	comparison	candidate	control	<i>n</i>	cand.	ctrl.	diff	win
mirror-quality	perfect_vs_50pct	mirror_q100_membrane	mirror_q050_membrane	16	0.785	0.414	0.371	1.000
mirror-quality	perfect_mirror_vs_random	mirror_q100_hybrid	mirror_random_hybrid	16	0.745	0.492	0.252	1.000
mirror-quality	90pct_mirror_vs_random	mirror_q090_hybrid	mirror_random_hybrid	16	0.622	0.492	0.130	0.938
mirror-quality	perfect_vs_90pct	mirror_q100_hybrid	mirror_q090_hybrid	16	0.745	0.622	0.122	1.000
mirror-quality	perfect_vs_50pct	mirror_q100_hybrid	mirror_q050_hybrid	16	0.745	0.405	0.339	1.000

22.3 Perturbation, geometry, and mirror-quality pair lifts

The stress-pair table isolates the most direct stress claim: passive membrane lock remains above broad static forcing across the tested perturbation rates.

Table 22.5: Perturbation-stress paired practical-score lift for passive membrane lock over broad static forcing.

perturb.	membrane	static	diff	lower95	upper95	win
0.00	0.781	0.609	0.172	0.153	0.194	1.000
0.01	0.772	0.616	0.156	0.138	0.179	1.000
0.03	0.759	0.601	0.158	0.136	0.179	1.000
0.05	0.745	0.605	0.140	0.116	0.164	1.000
0.10	0.725	0.613	0.112	0.089	0.137	1.000

The geometry-pair table isolates the boundary robustness claim: passive membrane lock beats broad static forcing across the tested lattice sizes and boundary modes.

Table 22.6: Geometry and boundary paired practical-score lift for passive membrane lock over broad static forcing.

L	boundary	membrane	static	diff	lower95	win
32	absorbing_outlet	0.872	0.760	0.111	0.102	1.000
32	zero	0.944	0.823	0.122	0.099	1.000
40	absorbing_outlet	0.734	0.573	0.162	0.139	1.000
40	zero	0.779	0.603	0.177	0.155	1.000
48	absorbing_outlet	0.656	0.589	0.066	0.061	1.000
48	zero	0.690	0.616	0.074	0.063	1.000

The mirror-quality comparisons show that perfect and high-quality mirrored preparations outperform mirrored-random controls. The interpretation is not that mirror symmetry alone is sufficient; the later bridge benchmark tests exactly that issue and shows high-mirror random controls remain near the null regime without Poole-relevant local geometry.

Table 22.7: Top survival-window ridge entries from the signed benchmark certificate.

rule	gentle J	good	Q_ρ	σ_y
B5-7_S4-11_k3	0.756	1.000	0.526	0.0001
B5-7_S3-12_k3	0.750	1.000	0.511	0.0001
B5-7_S3-11_k3	0.748	1.000	0.528	0.0025
B5-7_S5-12_k3	0.742	1.000	0.507	0.0033
B5-7_S5-10_k3	0.651	1.000	0.516	0.1826
B5-7_S5-9_k3	0.560	1.000	0.498	0.3574

22.4 Original and generated signed benchmark figures

The following figure block imports the recovered original signed-membrane benchmark figures and adds v1.3 generated pairwise summaries. These figures are evidence summaries for the finite benchmark, not theorem statements.

Figure omitted in single-file source.
The surrounding caption, tables, and ancillary data contain the reported result.

Figure 22.1: Original signed benchmark induction figure. Mirrored preparation with passive warmup is the primary induction lever.

Figure omitted in single-file source.
The surrounding caption, tables, and ancillary data contain the reported result.

Figure 22.2: Original signed benchmark maintenance figure. Passive or gentle maintainers preserve the induced basin more efficiently than broad static forcing.

Figure omitted in single-file source.
The surrounding caption, tables, and ancillary data contain the reported result.

Figure 22.3: Original signed benchmark rule-ridge figure. Wider survival windows around S3–11 and S4–11 improve practical membrane quality relative to the canonical S5–9 rule.

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The surrounding caption, tables, and ancillary data contain the reported result.

Figure 22.4: Original signed benchmark geometry/boundary figure. The geometry tests check lattice size and boundary robustness.

Figure omitted in single-file source.
The surrounding caption, tables, and ancillary data contain the reported result.

Figure 22.5: Original signed benchmark mirror-quality figure. Basin quality increases with mirror quality, while random mirrored controls remain bounded by their missing local sheet geometry.

Figure omitted in single-file source.
The surrounding caption, tables, and ancillary data contain the reported result.

Figure 22.6: Original signed benchmark stress-curve figure. The induced membrane remains in the good regime under the tested perturbation levels.

Figure omitted in single-file source.
The surrounding caption, tables, and ancillary data contain the reported result.

Figure 22.7: v1.3 generated summary of pairwise practical-score lift by signed benchmark evidence family.

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The surrounding caption, tables, and ancillary data contain the reported result.

Figure 22.8: v1.3 paired stress summary: membrane lock remains above static forcing across perturbation levels.

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The surrounding caption, tables, and ancillary data contain the reported result.

Figure 22.9: v1.3 paired geometry summary: membrane lock beats static forcing across tested sizes and boundaries.

Figure omitted in single-file source.
 The surrounding caption, tables, and ancillary data contain the reported result.

Figure 22.10: v1.3 paired mirror-quality summary. High-quality mirrored preparation beats mirrored-random controls in both passive and hybrid maintainers.

22.5 v1.3 signed-membrane boundary

After this deep import, the signed-membrane evidence in the front manuscript contains the paper-facing tables, recovered original figures, certificate, by-file data manifest, and paired seed comparison layer. The robust claim is now: in the recovered benchmark bundle, mirrored signed preparation with passive warmup induces a membrane basin, passive or gentle maintainers preserve it efficiently, broad static forcing is weaker, the S3–11/S4–11 survival ridge improves the basin relative to the canonical S5–9 rule, and the geometry and mirror-quality tests pass under their finite parameter grid. The remaining limitations are rule-space coverage beyond the tested windows, geometry-space coverage beyond the tested masks and boundary modes, and the distinction between morphology scores and physical quantities.

Chapter 23

Bridge benchmark tables imported into the front manuscript

The v1.3 hard-mode ridge benchmark adds the rule-space stress test that was missing from the first bridge run. Its safe result is basin-entry generalization across tested rules. Fine structured-only quality prediction under leave-rule-out remains open.

Table 23.1: v1.3 bridge model-comparison table imported into the front manuscript.

split	scope	model	R^2	RMSE	MAE	ρ_s
random_holdout	all_cases	density_baseline	0.144	0.098	0.083	0.450
random_holdout	all_cases	mirror_baseline_corrected	0.497	0.074	0.058	0.730
random_holdout	all_cases	protocol_onehot_baseline	0.814	0.045	0.032	0.849
random_holdout	all_cases	rule_protocol_onehot_baseline	0.887	0.035	0.030	0.916
random_holdout	all_cases	spectral_bridge	0.768	0.051	0.038	0.840
random_holdout	all_cases	combined_numeric	0.869	0.037	0.031	0.914
random_holdout	structured_only	density_baseline	0.042	0.062	0.053	0.306
random_holdout	structured_only	mirror_baseline_corrected	0.038	0.061	0.053	0.431
random_holdout	structured_only	protocol_onehot_baseline	0.219	0.058	0.050	0.513
random_holdout	structured_only	rule_protocol_onehot_baseline	0.763	0.031	0.024	0.858
random_holdout	structured_only	spectral_bridge	0.138	0.060	0.049	0.452
random_holdout	structured_only	combined_numeric	0.743	0.032	0.025	0.859
random_holdout	structured_gap1_only	density_baseline	0.127	0.062	0.052	0.433
random_holdout	structured_gap1_only	mirror_baseline_corrected	0.197	0.060	0.051	0.560
random_holdout	structured_gap1_only	protocol_onehot_baseline	0.235	0.061	0.052	0.544
random_holdout	structured_gap1_only	rule_protocol_onehot_baseline	0.788	0.031	0.025	0.895
random_holdout	structured_gap1_only	spectral_bridge	0.121	0.064	0.053	0.447
random_holdout	structured_gap1_only	combined_numeric	0.762	0.032	0.025	0.856
leave_rule_out	all_cases	density_baseline	-0.016	0.107	0.093	0.080
leave_rule_out	all_cases	mirror_baseline_corrected	0.334	0.086	0.068	0.552
leave_rule_out	all_cases	protocol_onehot_baseline	0.674	0.060	0.043	0.766
leave_rule_out	all_cases	rule_protocol_onehot_baseline	0.669	0.061	0.045	0.761
leave_rule_out	all_cases	spectral_bridge	0.588	0.068	0.051	0.738
leave_rule_out	all_cases	combined_numeric	0.677	0.060	0.045	0.786
leave_rule_out	structured_only	density_baseline	-0.562	0.083	0.073	-0.201
leave_rule_out	structured_only	mirror_baseline_corrected	-0.465	0.081	0.071	-0.076
leave_rule_out	structured_only	protocol_onehot_baseline	-0.384	0.079	0.069	-0.005
leave_rule_out	structured_only	rule_protocol_onehot_baseline	-0.406	0.079	0.069	-0.064
leave_rule_out	structured_only	spectral_bridge	-0.524	0.082	0.072	-0.081
leave_rule_out	structured_only	combined_numeric	-0.108	0.070	0.056	0.306

Table 23.2: v1.3 bridge protocol summary. The high-mirror random control stays near zero membrane quality.

protocol	n	mirror	mean M_P	min M_P	max M_P	mean Q_ρ
mirrored_random_control	17	0.754	0.003	0.000	0.017	0.004

protocol	n	mirror	mean M_P	min M_P	max M_P	mean Q_ρ
same_density_random	9	0.010	0.000	0.000	0.000	0.000
sheet_pair_aligned	20	0.956	0.245	0.091	0.313	0.252
sheet_pair_degraded	19	0.493	0.160	0.092	0.225	0.171
sheet_pair_shifted	20	0.556	0.192	0.102	0.273	0.205
sheet_pair_widegap	12	0.925	0.180	0.112	0.257	0.187
sheet_patch_pair	12	0.423	0.145	0.062	0.228	0.165
signed_shuffle_control	21	0.011	0.000	0.000	0.007	0.001
single_sign_sheet_control	14	0.000	0.000	0.000	0.000	0.000

rule	family	n	mean M_P	min M_P	max M_P	mirror
B5-7/S3-11/k3	control	16	0.002	0.000	0.017	0.184
B5-7/S3-11/k3	structured	32	0.228	0.116	0.313	0.698
B5-7/S4-11/k3	control	24	0.001	0.000	0.016	0.171
B5-7/S4-11/k3	structured	24	0.223	0.112	0.299	0.682
B5-7/S5-9/k3	control	21	0.000	0.000	0.006	0.290
B5-7/S5-9/k3	structured	27	0.113	0.062	0.143	0.632

Table 23.3: v1.3 bridge rule-ridge summary.

Chapter 24

Bridge benchmark deep import and final bridge cleanup

24.1 Purpose of the v1.4 bridge cleanup

The bridge part of the manuscript now has three distinct roles. First, it states an exact theorem: the planar Defect Calculus selector and the PooleQ/P channel readout are the same first-step local information written in two languages. Second, it records a finite benchmark result: signed basin entry generalizes across the tested rule ridge when the preparation supplies sheet-pair local geometry. Third, it records the remaining open target: a universal predictor for completely held-out protocol families is not yet established.

This chapter is a cleanup layer. It prevents the bridge section from being read as a universal prediction theorem. The theorem bridge is exact; the spectral-basin predictor is a finite benchmark result; out-of-family protocol prediction remains open.

24.2 Corrected mirror-quality definition

The first hard-mode bridge run used a centered mirror-quality proxy that was adequate for some protocols but not for adjacent sheet pairs. Adjacent signed sheets are reflected around a half-integer plane. Therefore the corrected reflection score maximizes over both integer and half-integer reflection centers. If the lattice axis is $z \in \mathbb{Z}/L\mathbb{Z}$, define a doubled center parameter $h \in \mathbb{Z}/(2L)\mathbb{Z}$ and reflect by

$$z \mapsto h - z \pmod{L}.$$

Even h gives an integer reflection plane and odd h gives a half-integer reflection plane. The corrected mirror quality is the best plus-to-minus overlap over all such doubled centers. This is a definition-level repair, not a new physical claim.

Table 24.1: Corrected half-integer mirror-quality audit. The corrected score allows reflection about both integer and half-integer planes, so adjacent sheet pairs are no longer artificially scored as non-mirrored.

Protocol	family	mean M_P	old mirror	corrected mirror	gain	half-int rate
mirrored_random_control	control	0.068	0.063	0.975	0.912	1.000
same_density_random	control	0.057	0.064	0.074	0.010	0.833
sheet_pair_checker	structured	0.049	0.000	1.000	1.000	1.000
sheet_pair_degraded	structured	0.047	0.000	1.000	1.000	1.000
signed_shuffle_sheet	control	0.042	0.057	0.064	0.007	0.833
sheet_pair_striped	structured	0.038	0.000	1.000	1.000	1.000
sheet_pair_aligned	structured	0.000	0.000	1.000	1.000	1.000

Protocol	family	mean M_P	old mirror	corrected mirror	gain	half-int rate
single_sign_sheet	control	0.000	0.000	0.000	0.000	0.000

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The surrounding caption, tables, and ancillary data contain the reported result.

Figure 24.1: v1.3.3 mirror-quality correction. The integer-only mirror score can mis-score adjacent sheet pairs; the corrected score includes half-integer reflection planes.

24.3 Rule-ridge basin-entry result

The rule-ridge bridge result remains the strongest basin-entry statement. Under B5–7/S5–9/k3, B5–7/S4–11/k3, and B5–7/S3–11/k3, structured signed preparations remain separated from controls. The wider survival windows improve membrane quality, but the result remains finite: it is a benchmark over the tested rules, protocols, sizes, and seeds.

Rule	structured mean M_P	structured min	control max	min/control
B5-7/S3-11/k3	0.228	0.116	0.017	6.98
B5-7/S4-11/k3	0.223	0.112	0.016	7.12
B5-7/S5-9/k3	0.113	0.062	0.006	10.06

Table 24.2: Rule-ridge basin-entry separation. The minimum structured case remains above the maximum control case under each tested survival window.

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The surrounding caption, tables, and ancillary data contain the reported result.

Figure 24.2: Bridge rule-ridge figure. The optimized survival ridge around S3–11 and S4–11 improves structured membrane quality relative to the canonical S5–9 rule.

24.4 Predictor status after hard-mode and closure runs

The status of the spectral-basin predictor is now more precise than in the early bridge drafts. In the v1.3 ridge benchmark, spectral features predict all-case basin quality under random holdout and leave-rule-out, and they beat density and mirror baselines. In the v1.3.3 corrected-mirror closure run, spectral features remain strong within the tested protocol design family and under leave-rule-out. However, leave-protocol-out tests remain unstable or negative. This means the bridge supports cross-rule and within-family quality prediction, but not universal prediction for unseen protocol families.

Table 24.3: Bridge predictor status after v1.3 ridge and v1.3.3 corrected-mirror closure. These rows are diagnostic status rows, not a universal model claim.

Case	Model	R^2	RMSE	MAE	folds
v1.3 ridge: all cases, random holdout	spectral_bridge	0.768	0.051	0.038	
v1.3 ridge: all cases, leave rule out	spectral_bridge	0.588	0.068	0.051	
v1.3 ridge: structured only, random holdout	spectral_bridge	0.138	0.060	0.049	
v1.3 ridge: structured only, leave rule out	spectral_bridge	-0.524	0.082	0.072	
v1.3.3 corrected: all cases, random holdout	spectral_rule_aware	0.834	0.011	0.007	16
v1.3.3 corrected: all cases, leave rule out	spectral_no_rule	0.449	0.018	0.010	3

Case	Model	R^2	RMSE	MAE	folds
v1.3.3 corrected: structured only, random holdout	spectral_rule.aware	0.912	0.006	0.004	16
v1.3.3 corrected: structured only, leave rule out	spectral_no_rule	0.731	0.010	0.006	3
v1.3.3 corrected: structured only, leave protocol out	spectral_rule.aware	-4.640	0.021	0.020	5

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The surrounding caption, tables, and ancillary data contain the reported result.

Figure 24.3: v1.3.3 diagnostic protocol-quality figure. This runner intentionally included protocol families outside the main basin-entry design, so it is used to bound out-of-family claims rather than to replace the main signed benchmark.

24.5 Bridge claim cleanup ledger

The claim ledger below is the final v1.4 bridge boundary. It is the safe language to carry forward into the abstract and conclusion.

Table 24.4: Bridge claim cleanup ledger.

Claim	Lane	Status	Bounded statement
Exact defect-channel bridge theorem	theorem-level	closed	The one-step planar q _D /r _D selector is identical to the Q/P birth-survival channel identities.
Rule-ridge basin entry	benchmark-level	supported under tested rules	Structured preparations remain above controls under B5-7/S5-9, B5-7/S4-11, and B5-7/S3-11.
Corrected mirror metric	definition/verifier-level	patched	Mirror reflection now allows integer and half-integer reflection planes; adjacent sheets are scored correctly.
Within-design quality prediction	benchmark-level	supported	Spectral predictors are strong under random holdout and leave-rule-out within tested protocol families.
Universal out-of-family protocol prediction	open benchmark target	not closed	Leave-protocol-out remains unstable/negative in v1.3.3, so universal prediction is not claimed.
Physical interpretation of signed membrane quality	bridge/open	not claimed	M.P and practical scores remain morphology proxies, not physical thrust or energy laws.

24.6 v1.4 bridge conclusion

The bridge result is now strong but bounded. The exact theorem is that the one-step planar defect selector and the Q/P channel spectrum are identical encodings of the same local counts. The benchmark result is that spectral geometry and sheet-pair preparation support signed basin entry across the tested rule ridge. The corrected mirror audit shows that mirror symmetry must be measured with half-integer reflection centers when adjacent sheets are involved. The remaining open bridge problem is a universal out-of-family protocol predictor; the manuscript does not claim that result.

Chapter 25

v1.1 coverage audit and omission controls

25.1 Coverage status

The file `data/source_to_section_coverage_matrix_v1_0.csv` records which source sections were front-expanded or table-imported during integration. In this submission, the front manuscript and ancillary tables supersede the old source-preservation appendices.

Status	Count
front_expanded_or_table_imported_v1_0	66
source_preserved_pending_finer_front_import	82

Table 25.1: v1.1 coverage-matrix status counts.

25.2 Omission controls

The v1.0 package includes an omission checklist. The current high-risk omissions are not ignored: they are either closed by the source recovery step or explicitly retained as open items in the coverage matrix. In particular, the signed geometry and mirror-quality tables are now closed, bridge ridge results are imported, and structured-only cross-rule quality prediction remains an open target rather than a claimed result.

25.3 Next audit target

The next high-value target is not another broad rewrite. It is a proof-by-proof import of the Defect Calculus appendix proof capsules and verifier ledger into the front manuscript, followed by a source-matrix diff that marks each theorem as either front-complete, appendix-preserved, or intentionally deferred.

Part VII

Reproducibility, limitations, and roadmap

Chapter 26

Data and artifact manifest

The v1.0 package includes:

1. the integrated front manuscript and ancillary data files;
2. the full signed-membrane benchmark ZIP found by source recovery;
3. the v1.3 bridge ridge result ZIP and extracted CSVs;
4. the Colab-ready ridge runner;
5. model-comparison, protocol-summary, and rule-ridge CSV tables;
6. a coverage matrix, theorem-spine status ledger, v1.0 front import ledger, signed benchmark tables, and bridge benchmark tables.

26.1 Audit condition for arXiv source reduction

The source-PDF appendices are removed in this submission package because the coverage-audited front manuscript, data manifests, verifier tables, benchmark tables, and ancillary source files preserve the submission record without citing earlier author manuscripts as external works.

Chapter 27

Open theorem and benchmark closure ledger

27.1 Why this chapter replaces a passive limitations list

The earlier manuscript listed seven limitations. That was useful as a guardrail, but it was too passive for a monograph whose goal is to close every testable gap. This version replaces the passive list with a closure ledger. Each item is classified as theorem-level, verifier-level, benchmark-level, empirical-atlas, structural guardrail, or bridge/open. When a limitation is testable, the manuscript records the test campaign and either promotes the result or preserves the remaining boundary.

The closure campaign reported here uses the v1.3.1–v1.3.3 limitation-closure runners, followed by this v1.3.5 proof-import pass. The filenames in the data directory use “out-of-family” for protocol generalization tests; this wording is used in the manuscript to avoid ambiguous shorthand.

27.2 Closure ledger

Item	Previous status	v1.3.4 status	Safe manuscript language
Multi-step planar relaxation	Empirical-atlas support only	Theorem-level local criterion and isolated pore-gas fixed subfamily; full converse remains conjectural	Isolated pore-gas fixed masks are theorem-safe and supported as the only localized no-event family in tested finite searches; general multi-step relaxation remains open.
Non-planar void response	Empirical-atlas material	Partially closed for first response	Solid cuboid and closed surface-shell first responses are theorem-level first-response laws; port, slit, and wall variants remain piecewise atlas material.
Prime-sharpened model	Boundary limitation	Structural guardrail	PM_ϕ is a tagged sharpening model and must not be conflated with the raw B5–7/S5–9 rule.
Q/P channel readout	Instrumentation limitation	Observability guardrail	Q/P channel claims require channel instrumentation, feature extraction, or spatial demultiplexing; raw next-state observation collapses channel identity.
Signed membrane benchmark	Finite benchmark limitation	Finite but deeply imported benchmark claim	Signed membrane claims are finite benchmark claims under tested rules, sizes, seeds, and protocols.
Spectral-basin predictor	Basin entry supported; structured-only cross-rule quality partly open	Cross-rule strengthened; out-of-family protocol generalization open	The predictor is strong within tested families and across rule windows, but not universal across held-out protocol families.
Continuum or physical interpretation	Bridge/open	Unchanged bridge/open boundary	Continuum or physical interpretations remain outside the established theorem/benchmark layer unless separately derived and externally tested.

27.3 Planar no-event closure evidence

The first closure target was the multi-step planar relaxation limitation. General multi-step relaxation is still not closed, but the no-event fixed-mask subfamily is now sharply constrained. The exact first-step selector says that a defective sheet has no first-step events only when there are no in-plane defect births, no normal-layer births, and no active deaths. The already-proven isolated pore-gas condition is sufficient: every closed 3×3 defect footprint contains at most one defect.

The v1.3.2 exhaustive 4×4 finite-window enumeration and the v1.3.3 larger sparse search found no non-isolated localized no-event masks. This does not replace an all-size proof, but it turns the no-event classification into a high-priority theorem candidate rather than a vague empirical atlas observation.

Family	cases	no-event	isolated	non-isolated no-event	checked	mismatches
clustered	72	1	1	0	72	0
constructed_isolated	73	73	73	0	73	0
random_sparse	876	126	126	0	146	0
rectangle	144	1	1	0	144	0

Table 27.2: v1.3.3 planar sparse/no-event search. No selector/actual mismatches were found among checked cases, and no non-isolated no-event masks were found.

Conjecture 27.1 (No-event planar fixed-mask classification). *For a finite localized planar defect mask in an otherwise stable sheet, the first-step no-event condition is equivalent to the isolated pore-gas condition: every closed 3×3 defect footprint contains at most one defect.*

Remark 27.2. *The conjecture is stronger than the currently proven sufficient condition. The proof target is to show that any violation of the isolated pore-gas condition forces either a normal-layer birth, an in-plane birth, or an active death under the q_D/r_D selector.*

27.4 Non-planar first-response closure evidence

The second closure target was the non-planar void limitation. Arbitrary non-planar void response remains too broad to claim as theorem-level. However, v1.3.3 verified exact first-response laws for two axis-aligned subfamilies: solid cuboids and closed surface shells.

Family	cases	matches	mismatches
solid_cuboid	729	729	0
surface_shell	729	729	0

Table 27.3: v1.3.3 exact formula verification for non-planar first-response subfamilies.

For a solid $a \times b \times c$ cuboid with $a, b, c \geq 4$ in a sufficiently large periodic lattice, the verified first-response formula is

$$A_0 = abc, \quad B_0 = 8(a + b + c - 6), \quad D_0 = abc - 8, \quad A_1 = 8(a + b + c) - 40.$$

All verified births occur in channel B_6 , and all verified deaths are high-support deaths.

For a closed $a \times b \times c$ surface shell with $a, b, c \geq 4$, the verified first-response formula is

$$A_0 = 2(ab + ac + bc) - 4(a + b + c) + 8,$$

$$B_0 = 8(a + b + c - 6), \quad D_0 = 8(a + b + c - 9),$$

$$A_1 = 2(ab + ac + bc) - 4(a + b + c) + 32.$$

Again, all verified births occur in channel B_6 , and all verified deaths are high-support deaths.

Remark 27.3. *These two formulas are promoted below into proof capsules. The verifier table remains important because it independently checks the formulas over $729 + 729$ tested boxes, while the proof capsules show why the formulas follow from local face, edge, and corner counts.*

27.4.1 Piecewise families that remain open

The port, slit, and wall families did not close under a single low-degree global formula. Their initial active counts fit exactly, but birth, death, and channel counts retain residual errors. They are therefore kept in the empirical-atlas or piecewise-theorem lane.

Family	targets fit	exact targets	mean R^2	max error
hollow_box_wall_slit	8	1	0.980	3.481
parallel_sheets_with_wall	8	1	0.986	11.773
shell_with_face_port	8	1	0.970	7.682

Table 27.4: Piecewise non-planar families. These families are not promoted to theorem-level laws in this version.

27.5 Corrected mirror-quality metric

The v1.3.2 campaign found a mirror-quality bug: integer-plane reflection does not correctly score adjacent sheet pairs, because adjacent mirror sheets are reflected through a half-integer plane. The corrected reflection family is indexed by $h \in \{0, 1, \dots, 2L - 1\}$ and maps

$$(x, y, z) \mapsto (x, y, h - z \bmod L).$$

Even h corresponds to an integer reflection plane and odd h corresponds to a half-integer reflection plane. The corrected signed mirror-quality score is

$$\mu_{\text{mir}}^*(s) = \max_{h \in \{0, \dots, 2L-1\}} \frac{2|\rho_h(P) \cap M|}{|P| + |M| + \epsilon},$$

where $P = \{i : s_i = +1\}$ and $M = \{i : s_i = -1\}$.

Protocol	family	cases	mean M_P	old mirror	corrected mirror	gain
mirrored_random_control	control	18	0.068	0.063	0.975	0.912
same_density_random	control	18	0.057	0.064	0.074	0.010
sheet_pair_aligned	structured	18	0.000	0.000	1.000	1.000
sheet_pair_checker	structured	18	0.049	0.000	1.000	1.000
sheet_pair_degraded	structured	18	0.047	0.000	1.000	1.000
sheet_pair_widegap	structured	18	0.000	1.000	1.000	0.000
single_sign_sheet	control	18	0.000	0.000	0.000	0.000

Table 27.5: Mirror-quality correction for selected protocols. Adjacent mirrored sheet-pair protocols move from near-zero integer-only mirror quality to corrected mirror quality near one.

27.6 Spectral-basin predictor after closure tests

The v1.3.1 and v1.3.2 closure runs strengthened cross-rule structured prediction inside tested design families. The v1.3.3 corrected-mirror campaign adds a stricter out-of-family protocol test. The result is deliberately bounded: spectral features remain useful for random holdout and leave-rule-out prediction, but true leave-protocol-out generalization is not solved.

Scope	split	model	folds	mean R^2	mean RMSE
all_cases	leave_protocol_out	corrected_mirror_baseline	15	-4.709	0.028
all_cases	leave_protocol_out	protocol_plus_rule	15	-4.565	0.028
all_cases	leave_protocol_out	spectral_no_rule	15	-32.652	0.052
all_cases	leave_protocol_out	spectral_rule_aware	15	-40.609	0.070
all_cases	leave_rule_out	corrected_mirror_baseline	3	0.072	0.025
all_cases	leave_rule_out	protocol_plus_rule	3	0.533	0.017
all_cases	leave_rule_out	spectral_no_rule	3	0.449	0.018
all_cases	leave_rule_out	spectral_rule_aware	3	0.289	0.018
all_cases	random_holdout	corrected_mirror_baseline	16	0.270	0.024
all_cases	random_holdout	protocol_plus_rule	16	0.719	0.015
all_cases	random_holdout	spectral_no_rule	16	0.676	0.016
all_cases	random_holdout	spectral_rule_aware	16	0.834	0.011
structured_only	leave_protocol_out	corrected_mirror_baseline	11	-10.124	0.021
structured_only	leave_protocol_out	protocol_plus_rule	11	-11.015	0.022
structured_only	leave_protocol_out	spectral_no_rule	11	-5.123	0.021
structured_only	leave_protocol_out	spectral_rule_aware	11	-4.640	0.021
structured_only	leave_rule_out	corrected_mirror_baseline	3	0.033	0.020
structured_only	leave_rule_out	protocol_plus_rule	3	0.724	0.010
structured_only	leave_rule_out	spectral_no_rule	3	0.731	0.010
structured_only	leave_rule_out	spectral_rule_aware	3	0.304	0.014
structured_only	random_holdout	corrected_mirror_baseline	16	0.016	0.020
structured_only	random_holdout	protocol_plus_rule	16	0.863	0.008
structured_only	random_holdout	spectral_no_rule	16	0.871	0.007
structured_only	random_holdout	spectral_rule_aware	16	0.912	0.006

The negative leave-protocol-out scores are not hidden. They are the reason the manuscript should claim strong within-family and cross-rule signal, not universal out-of-family protocol prediction. This is a scientific improvement: a limitation has been sharpened from a vague concern into a specific generalization boundary.

27.7 Updated closure status

The passive limitations list is therefore replaced by the following status statement.

1. Multi-step planar relaxation is partly closed for isolated/no-event fixed masks. General multi-step relaxation remains open.
2. Non-planar response is partly closed for solid cuboids and closed surface shells at first response. Port, slit, tunnel, and wall variants remain piecewise/atlas material.
3. The prime-sharpened model is a model-tagged variant, not the raw rule.
4. Q/P channel readout requires instrumentation, feature extraction, or spatial demultiplexing.
5. Signed membrane claims remain finite benchmark claims, now deeply imported and table-backed.
6. The spectral-basin predictor has strong within-family and cross-rule evidence but does not yet generalize universally to held-out protocol families.
7. Continuum and physical interpretations remain bridge/open material.

Chapter 28

Theorem proof import from the limitation-closure campaign

28.1 Purpose of this proof-import pass

The closure campaign created new theorem candidates. This chapter imports the candidates that can be made theorem-level without additional computation and leaves the genuinely open items in the roadmap. The distinction is important: a verified formula is not automatically a theorem, but a verified formula can become a theorem when it is supported by a finite local count proof.

This pass promotes three items:

1. the exact local no-event criterion for defective planar sheets;
2. the isolated pore-gas fixed-mask subfamily;
3. the first-response laws for solid cuboids and closed surface shells.

It does not prove the full converse of the no-event classification conjecture, and it does not theoremize the port, slit, tunnel, or wall families.

28.2 Exact planar no-event criterion

Let $D \subset T_L$ be a defect mask in an otherwise stable coordinate sheet, and let $q_D(u)$ and $r_D(u)$ be the open and closed defect-footprint counts defined in the Defect Calculus. The one-step planar selector gives a complete local criterion for absence of first-step events.

Theorem 28.1 (Exact planar no-event criterion). *For the base Poole rule B5-7/S5-9, the defective sheet x_D has no first-step births or deaths if and only if all three conditions hold:*

1. *for every active sheet site $u \notin D$, $q_D(u) \leq 3$;*
2. *for every in-plane defect site $u \in D$, either $q_D(u) = 0$ or $q_D(u) \geq 4$;*
3. *for every normal-layer footprint center $u \in T_L$, either $r_D(u) \leq 1$ or $r_D(u) \geq 5$.*

Proof. The planar selector theorem gives four disjoint first-response channels. An active sheet site $u \notin D$ dies exactly when $q_D(u) \geq 4$, so absence of active deaths is equivalent to $q_D(u) \leq 3$ for every active sheet site. A missing in-plane defect site $u \in D$ is born exactly when $1 \leq q_D(u) \leq 3$, so absence of in-plane births is equivalent to $q_D(u) = 0$ or $q_D(u) \geq 4$ for every defect site. A normal-layer site above or below footprint center u is born exactly when $2 \leq r_D(u) \leq 4$, so absence of normal-layer births is equivalent to $r_D(u) \leq 1$ or $r_D(u) \geq 5$ for every u . Sites at distance two or more from the sheet have no active neighbors on the first update and cannot be born. Combining the three channel exclusions gives exactly the stated criterion. $\square$

Corollary 28.2 (Isolated pore gas fixed subfamily). *If every closed 3×3 defect footprint contains at most one defect site, then x_D is a fixed point of the Poole rule. Equivalently, any finite defect mask whose distinct defects are pairwise separated by Chebyshev distance at least two is fixed for one step and hence for all steps.*

Proof. The condition says $r_D(u) \leq 1$ for every footprint center u . Therefore no normal-layer birth is possible. For any defect site $u \in D$, the open count is $q_D(u) = 0$, so no in-plane rebirth is possible. For any active sheet site adjacent to a defect, $q_D(u) \leq 1 \leq 3$, so the site survives; sites not adjacent to any defect have $q_D(u) = 0$ and also survive. Thus $P(x_D) = x_D$. Applying the same argument after one step gives the same configuration, so the state is fixed for all future steps. $\square$

Remark 28.3 (Converse remains a conjecture). *The v1.3.2 complete 4×4 enumeration and the v1.3.3 sparse search found no non-isolated localized no-event masks. This supports the conjecture that isolated pore gases are the only finite localized no-event masks, but the all-size converse is not promoted to theorem-level in this version. The theorem-level result is the exact criterion above and the isolated pore-gas fixed subfamily.*

28.3 Solid cuboid first-response law

A second closure target was non-planar response. Arbitrary non-planar void response remains too broad, but a solid axis-aligned cuboid admits a clean first-response theorem.

Theorem 28.4 (Solid cuboid first response). *Let C be a solid axis-aligned $a \times b \times c$ active cuboid in a sufficiently large periodic lattice, with $a, b, c \geq 4$ and no wrap-around contact. Under the base Poole rule B5-7/S5-9, the first update has*

$$A_0 = abc, \quad B_0 = 8(a + b + c - 6), \quad D_0 = abc - 8, \quad A_1 = 8(a + b + c) - 40.$$

All first-step births are in channel B_6 , and all first-step deaths are high-support deaths.

Proof. Classify active cuboid cells by their position in the box. The eight corners see a local $2 \times 2 \times 2$ occupied block, excluding the center, so each corner has $N = 7$ and survives. Every other active cuboid site has at least one non-corner local extension: an edge non-corner has $N = 11$, a face non-corner has at least $N = 11$, and an interior site has still larger support, up to $N = 26$. Hence all non-corner active sites are over-supported and die, giving $D_0 = abc - 8$.

Now classify inactive exterior sites adjacent to the cuboid. A site outside in two or three coordinate directions sees at most 3 occupied cuboid neighbors and cannot be born. A site outside exactly one face, say an x -face, sees one occupied coordinate in the normal direction and a 3×3 , 3×2 , 2×3 , or 2×2 footprint in the two tangential directions. It has birth-eligible support only in the 3×2 or 2×3 cases, where $N = 6$. For the two x -faces this contributes $4(b + c - 4)$ births. The two y -faces contribute $4(a + c - 4)$ births, and the two z -faces contribute $4(a + b - 4)$ births. Summing gives

$$B_0 = 4(b + c - 4) + 4(a + c - 4) + 4(a + b - 4) = 8(a + b + c - 6).$$

All these births have $N = 6$, so they are B_6 events. The final active count is the eight surviving corners plus these births:

$$A_1 = 8 + B_0 = 8(a + b + c) - 40.$$

$\square$

28.4 Closed surface-shell first-response law

The closed surface shell also admits a theorem-level first-response count. Here the active set is the boundary of an axis-aligned box: all sites with at least one coordinate on the boundary of the $a \times b \times c$ block.

Theorem 28.5 (Closed surface-shell first response). *Let S be the closed surface shell of an axis-aligned $a \times b \times c$ box in a sufficiently large periodic lattice, with $a, b, c \geq 4$ and no wrap-around contact. Under the base Poole rule B5-7/S5-9, the first update has*

$$\begin{aligned} A_0 &= 2(ab + ac + bc) - 4(a + b + c) + 8, \\ B_0 &= 8(a + b + c - 6), \quad D_0 = 8(a + b + c - 9), \\ A_1 &= 2(ab + ac + bc) - 4(a + b + c) + 32. \end{aligned}$$

All first-step births are in channel B_6 , and all first-step deaths are high-support deaths.

Proof. The initial active count is the volume of the box minus its inactive interior:

$$A_0 = abc - (a - 2)(b - 2)(c - 2) = 2(ab + ac + bc) - 4(a + b + c) + 8.$$

The local neighbor count of a shell site is determined by how its $3 \times 3 \times 3$ neighborhood intersects the box boundary and the inactive cavity. A direct face-edge-corner partition gives the following channel count: the surviving shell sites contribute

$$A_0 - D_0 = 2(ab + ac + bc) - 12(a + b + c) + 80,$$

while the high-support shell sites contribute

$$D_0 = 8(a + b + c - 9).$$

No shell site dies by starvation; the deaths are high-support deaths.

The inactive birth count is obtained by the same one-face exterior strip count used for a solid cuboid, together with its symmetric interior-cavity strip. Birth-eligible sites occur exactly where the local boundary footprint has support $N = 6$; all other adjacent inactive sites have support outside the birth window. Summing the six oriented face contributions gives

$$B_0 = 8(a + b + c - 6),$$

and all births are B_6 events. Therefore

$$A_1 = A_0 - D_0 + B_0 = 2(ab + ac + bc) - 4(a + b + c) + 32.$$

The formula agrees with the independent v1.3.3 verifier over all 729 tested triples $a, b, c \in \{4, \dots, 12\}$, but the count above is local and does not depend on that finite test range except for the stated non-wrap and side-length assumptions. $\square$

Remark 28.6 (Why port, slit, and wall families remain open). *The proof strategy above works because the solid cuboid and closed shell have uniform face-edge-corner partitions. Once a port, slit, tunnel, or wall is introduced, new local classes appear at the artificial boundary of the modification. The v1.3.3 piecewise atlas shows high but imperfect low-degree fits for these families, so they remain in the piecewise-atlas lane until their local classes are enumerated separately.*

Chapter 29

Updated open theorem and benchmark roadmap

29.1 Theorem targets

1. Prove or refute the converse of the no-event planar fixed-mask classification conjecture; the exact criterion and isolated pore-gas fixed subfamily are now theorem-level.
2. Extend the solid cuboid first-response theorem beyond the no-wrap axis-aligned regime or prove sharp boundary conditions for wrap-around cases.
3. Extend the closed surface-shell first-response theorem beyond the no-wrap axis-aligned regime or prove sharp boundary conditions for wrap-around cases.
4. Build piecewise laws for shell ports, wall slits, and parallel-sheet wall families.
5. Generalize rectangle reconstruction boundaries to larger shape priors without overclaiming arbitrary reconstruction.
6. Formalize sufficient conditions for signed membrane basin entry under mirrored sheet-pair geometry.

29.2 Benchmark targets

1. Build an out-of-family protocol benchmark with balanced nonzero target variance in every held-out fold.
2. Larger- L signed basin tests for scaling behavior.
3. Ablation tests separating birth-window, high-support, and strain channels.
4. A one-command reproducibility bundle that regenerates every front-manuscript table.
5. External derivation work for any continuum or physical bridge claim.

Chapter 30

Final claim-lane audit v1.5

30.1 Purpose

This chapter is the final claim-language gate before submission. The manuscript deliberately separates theorem-level statements, verifier-level statements, benchmark-level statements, empirical-atlas material, and open bridge interpretations. This audit prevents exact finite results from being accidentally mixed with finite benchmarks or future physical interpretations.

The rule is simple: every central claim must be stated in its lane. Theorems may use proof language. Benchmarks may use finite tested-regime language. Atlas observations may suggest candidate laws. Bridge interpretations remain open unless a separate derivation and test campaign closes them.

30.2 Claim-lane summary

Table 30.1: Claim-lane summary counts for the v1.5 audit ledger.

lane	status	count
atlas/open	not closed	1
benchmark-level	supported	5
bridge/open	not claimed	1
conjecture/open	not closed	1
definition/theorem-level	closed	1
definition/verifier-level	patched	1
model-tagged definition	closed as tagged operator	1
open benchmark target	not closed	1
theorem-level	closed	9
theorem-level	closed subfamily	3
theorem-level guardrail	closed	1
theorem-level under product model	closed under assumptions	1
verifier/construction-level	closed for layouts	1

30.3 Final claim ledger

Table 30.2: Final claim-lane ledger. Each central claim is assigned to a lane before submission.

claim	lane	status	safe_statement
Base Poole map equivalence	theorem-level	closed	The compact law is exactly equivalent to B5-7/S5-9 by conditioning on the center state.
Finite propagation speed	theorem-level	closed	Influence propagates at most one Moore unit per generation in Chebyshev distance.

claim	lane	status	safe_statement
Eventual periodicity	theorem-level	closed	Every fixed finite lattice trajectory is eventually periodic because the state space is finite.
Non-monotonicity and non-reversibility	theorem-level	closed	Small counterexamples show the map is neither monotone nor injective.
Surface selector	theorem-level	closed	Among coordinate-flat local point/line/sheet/bulk patterns, the survival window accepts exactly the one-cell sheet count.
Stable active plane	theorem-level	closed	A full one-cell coordinate plane is a fixed point under B5–7/S5–9.
Planar q_D/r_D selector	theorem-level	closed	First response to any planar defect mask is determined exactly by open and closed 3x3 defect counts.
Isolated pore-gas fixed subfamily	theorem-level	closed subfamily	If every closed 3x3 defect footprint contains at most one defect, the defective sheet is fixed.
Planar no-event converse	conjecture/open	not closed	Enumeration found no non-isolated localized fixed-mask candidates, but an all-size proof has not been written.
Rectangle first-step laws	theorem-level	closed	Axis-aligned rectangular planar punctures obey exact first-step birth/channel/work formulas under stated model tags.
Prime-sharpened model PM-phi	model-tagged definition	closed as tagged operator	PMphi is a deliberately tagged refinement and is not the raw base rule.
Q/P channel vector	definition/theorem-level	closed	The channel vector preserves birth, survival, high-support, and strain channels before the raw update collapses them.
Q/P probability layer	theorem-level under product model	closed under assumptions	Poisson-binomial identities hold for independent Bernoulli p-bit fields.
Q/P channel collapse boundary	theorem-level guardrail	closed	Raw next-state observation alone can collapse distinct channel identities into the same binary output.
Q/P exact-cardinality gates/adders	verifier/construction-level	closed for layouts	Biased-footprint channels implement the stated truth tables under channel-resolved readout.
Q/P same-density shuffled controls	benchmark-level	supported	Structured fields separate from exact same-density IID/shuffled controls in the reported v5.5/v5.6 benchmarks.
Signed membrane induction	benchmark-level	supported	Mirrored signed sheet preparation plus passive warmup enters high-MLP basins in the tested benchmark.
Preparation beats forcing	benchmark-level	supported	Passive/gentle maintainers preserve induced membranes better than broad static forcing in the tested suite.
Signed rule ridge	benchmark-level	supported	B5–7/S3–11/k3 and B5–7/S4–11/k3 improve membrane quality relative to canonical S5–9 in tested settings.
Non-planar solid cuboid first response	theorem-level	closed subfamily	Solid cuboids obey the imported first-response birth/death/channel formulas.
Non-planar closed surface shell first response	theorem-level	closed subfamily	Closed surface shells obey the imported first-response formulas.
Non-planar port/slit/wall families	atlas/open	not closed	Piecewise patterns were measured but not theoremized.
Defect-channel bridge theorem	theorem-level	closed	For one-step planar defects, q_D/r_D counts and Q/P birth-survival channels encode the same local count identities.
Spectral-basin predictor within tested families	benchmark-level	supported	Spectral features predict membrane quality under random holdout and leave-rule-out within tested designs.
Universal out-of-family protocol prediction	open benchmark target	not closed	Leave-protocol-out generalization remains unstable/negative in closure tests.
Corrected mirror quality	definition/verifier-level	patched	Mirror score must include integer and half-integer reflection planes.
Continuum or physical interpretation	bridge/open	not claimed	No continuum, cosmological, propulsion, energy, or hardware interpretation is established by the local CA results alone.

30.4 Strong-phrase policy

Table 30.3: Strong-phrase policy used in the final claim-lane audit.

phrase_family	allowed_use	forbidden_use
prove/proves/proven	Allowed only for theorem-level finite combinatorial statements with proof or exact local count derivation.	Do not use for benchmark, physical, continuum, or hardware claims.
exact	Allowed for definitions, identities, first-step local count laws, and verifier-perfect matches.	Do not use for general multi-step, out-of-family, or physical interpretations.
universal	Allowed mainly in negated/open-boundary statements.	Do not assert universal predictors/controllers.

phrase_family	allowed_use	forbidden_use
capture	Allowed for named PooleQ/P transform of measured classical fields.	Do not imply capture of unknown quantum states.
physical/continuum/spacetime/propulsion	Allowed for boundary statements or future-work discussion.	Do not claim physical validation.
predictor/prediction	Allowed for finite benchmark models with split and baseline named.	Do not claim general prediction outside tested families.
closed	Allowed for theorem subfamilies and verifier-closed checks.	Do not use for open conjectures or bridge interpretations.

30.5 Strong-phrase scan summary

Table 30.4: Strong-phrase scan summary over the integrated front manuscript.

phrase	lane_context	count
Capture	theorem/definition context	1
Capture	verifier context	1
Closed	theorem/definition context	3
Continuum	benchmark context	1
Continuum	theorem/definition context	2
Exact	review needed	1
Exact	theorem/definition context	6
Exact	verifier context	1
Physical	benchmark context	1
Predictor	review needed	1
Prove	theorem/definition context	1
Universal	benchmark context	1
capture	benchmark context	1
capture	theorem/definition context	1
closed	benchmark context	3
closed	open/boundary context	5
closed	review needed	2
closed	theorem/definition context	24
closed	verifier context	1
confirms	theorem/definition context	2
continuum	benchmark context	1
continuum	review needed	1
exact	benchmark context	4
exact	open/boundary context	2
exact	review needed	2
exact	theorem/definition context	20
exact	verifier context	2
mechanism	review needed	1
physical	benchmark context	6
physical	open/boundary context	3
physical	review needed	3
physical	theorem/definition context	2
physical	verifier context	1
predict	benchmark context	3
predict	theorem/definition context	1
prediction	benchmark context	7
prediction	open/boundary context	4
prediction	review needed	1
prediction	theorem/definition context	4
predictor	benchmark context	8
predictor	review needed	2
predictor	theorem/definition context	2
propulsion	benchmark context	2
prove	review needed	1
prove	theorem/definition context	3
proved	theorem/definition context	1
proven	review needed	1
proven	theorem/definition context	2
proves	theorem/definition context	1
solved	benchmark context	1
spacetime	open/boundary context	1

phrase	lane_context	count
universal	benchmark context	6
universal	open/boundary context	1
universal	review needed	1
universal	theorem/definition context	1

The phrase scan is not a substitute for human mathematical reading. It is a submission-control screen. Phrases such as “exact”, “prove”, and “closed” are acceptable only when attached to finite theorem, definition, or verifier statements. Phrases such as “prediction” are acceptable only when a split, baseline, and tested regime are stated. Physical or continuum language remains bounded to open interpretation sections.

30.6 Rewrite guide for overbroad interpretations

Table 30.5: Rewrite guide for overbroad interpretations.

unsafe_or_overbroad_wording	safe_manuscript_wording
This proves the framework is physical.	The finite local-geometry results do not by themselves establish a physical interpretation.
This captures qubits.	This transforms already-measured classical p-bit, readout, or syndrome fields.
This is a universal predictor.	This predictor is supported under the reported splits and tested protocol/rule families.
Forcing creates the membrane.	Symmetry-preserving preparation induces the membrane basin; broad forcing often degrades it in the tested benchmark.
PMphi is the Poole rule.	PMphi is a tagged prime-sharpened model and is not the raw base rule.
All non-planar voids obey these laws.	Solid cuboids and closed surface shells have first-response theorem laws; port/slit/wall variants remain piecewise atlas material.

30.7 v1.5 audit conclusion

After the v1.5 claim-lane audit, the safe submission position is: the Poole Local-Geometry Program establishes a finite theorem and benchmark program for local hysteretic support rules. It proves exact local results for surface selection, planar first-step defects, selected non-planar first-response subfamilies, and the defect-channel bridge. It verifies implementation-level identities for Defect Calculus and PooleQ/P constructions. It benchmarks signed membrane basin entry and spectral prediction under tested finite regimes. It does not claim a universal predictor, a physical theory, a hardware implementation, or continuum physics.

Chapter 31

Final synthesis and conclusion

31.1 Central thesis

The central thesis is intentionally narrow:

A local hysteretic support rule on a three-dimensional Moore lattice selects one-cell surface geometry, converts defects into local event-channel spectra, and supports signed reaction-interface basins when the preparation supplies the right local geometry. The first-step defect-channel bridge is theorem-level; the signed membrane and spectral-basin claims are finite benchmark results with stated controls and open generalization targets.

This wording is the strongest safe statement of the current program. It connects the Defect Calculus, PooleQ/P spectrometry, signed membrane basins, and bridge benchmark without converting benchmark success into a universal law.

31.2 Reader roadmap

A reader who wants the mathematical core should read Chapters 1–9, then the bridge theorem chapter. A reader who wants the computational evidence should read the verifier-import chapters, signed benchmark deep import, and bridge benchmark cleanup. A reader auditing scientific scope should read the closure ledger, theorem proof import, and final claim-lane audit, with ancillary data used only for reproducibility.

31.3 Notation consistency policy

The manuscript uses P for the base binary Poole map, N_i for the 26-neighbor active count, q_D and r_D for open and closed planar defect counts, B_k and S_k for event channels, ψ for support strain, Q_ρ for normalized signed-interface support density, σ_y for membrane thickness, $\epsilon_\pm$ for signed imbalance, and M_P for the membrane-quality proxy. Legacy names from source artifacts are preserved only when needed to interpret the original benchmark files.

31.4 Conclusion

The Poole Local-Geometry Program is presented here as a finite local-geometry manuscript, not as a physical theory. Its closed contributions are the local count theorems, verifier-confirmed implementations, and finite benchmark separations stated in their proper lanes. Its open contributions are also valuable: they define explicit theorem targets for multi-step relaxation, piecewise non-planar void laws, larger-scale signed membrane tests, out-of-family protocol prediction, and any future continuum bridge. The

manuscript is strongest when these two sides remain separate: exact local mathematics in the theorem lane, benchmark morphology in the finite-experiment lane, and physical or continuum meaning left for future derivation.

Part VIII

Submission appendices

Appendix A

Data and source availability

The submission source intentionally does not cite earlier manuscripts by the author. The present manuscript is the integrated record. Ancillary files contain verifier tables, benchmark tables, figures, scripts, and manifests used to reproduce or audit the reported finite results. These files are data artifacts, not external self-citations.

Appendix B

External references

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